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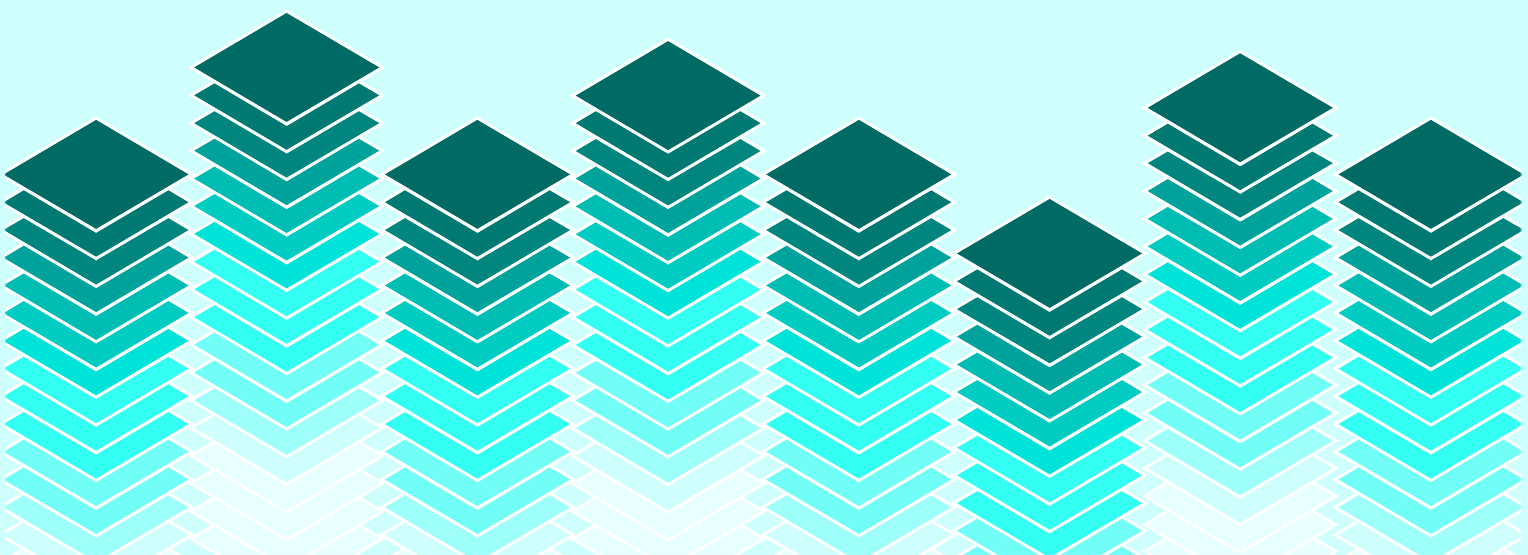
Federal Ministry for Economic Affairs and Energy

on the basis of a decision by the German Bundestag

Navigating the Digital Battery Passport

User Stories from a Technical Perspective

Version 1.0 / June 2026



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NAVIGATING THE DIGITAL BATTERY PASSPORT: USER STORIES FROM A TECHNICAL PERSPECTIVE

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GEFEG Passion for Standards

ABOUT THIS DOCUMENT

This document presents the results of a structured analysis of the Digital Product Passport (DPP) ecosystem conducted within the BatteryPass-Ready project, applying a technical perspective to the requirements arising from the Battery Regulation (Regulation (EU) 2023/1542), the Ecodesign for Sustainable Products Regulation (ESPR, Regulation (EU) 2024/1781), and the standardization activities of the CEN/CENELEC Joint Technical Committee 24 (JTC 24). It considers the six JTC 24 standards published in May 2026, as well as two draft standards published in July/August 2025 that are currently under approval and expected to progress further in Q3 2026. The analysis covers five core business processes within the battery passport lifecycle. Registering a battery and placing it on the market, accessing battery passport data, maintaining battery passport data, searching for battery passports, and managing end-of-life and circularity events. For each business process, one or more user stories are developed, detailing the involved actors and their roles, the ordinary technical sequence of interactions, and the failure and test scenarios that arise from them. The failure and test scenarios are expressed in natural language rather than as executable tests, making them accessible to both technical and non-technical readers alike. The test scenarios defined in this document serve as the foundation for the test environment currently being developed within the BatteryPass-Ready project.

The primary audience comprises organizations engaged in the development or integration of DPP systems, with a particular focus on small and medium-sized enterprises (SMEs). For these organizations, the user stories clarify what needs to be done, by whom, and in what order, providing a practical entry point into the technical requirements of battery passport compliance. For DPP system designers and technical implementers, the failure and test scenarios describe the critical interaction points that a robust DPP system must be able to handle correctly. Where the regulatory framework and applicable standards do not provide sufficient guidance, explicit assumptions based on internal discussions and stakeholder feedback have been made and are clearly marked throughout the document.

Disclaimer: This document has been prepared by the BatteryPass-Ready consortium and reflects the status of the regulatory and standardization landscape as of June 2026. It is intended to support organizations in understanding and navigating the technical requirements of the battery passport ecosystem and should not be considered exhaustive. As the regulatory framework and associated standardization activities addressed in this document are still evolving, the findings presented here may require revision as requirements are further specified, adopted, or updated. This document does not prescribe specific technical solutions, does not constitute legal advice, and does not express the official position of any regulatory authority.

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ABBREVIATIONS

Abbreviation	Meaning
BSP	Backup Service Provider
DBP	Digital Battery Passport
DPP	Digital Product Passport
EC	European Commission
EO	Economic Operator
ESPR	Ecodesign for Sustainable Products Regulation
EU	European Union
EV	Electric Vehicle
ID	Identifier
JTC 24	Joint Technical Committee 24
LMT	Light Means of Transport
MSA	Market Surveillance Authority
MSP	Main Service Provider
SME	Small and Medium-sized Enterprise
URI	Uniform Resource Identifier
URL	Uniform Resource Locator

1 INTRODUCTION

The Digital Product Passport (DPP) is a structured digital record that accompanies a product throughout its entire lifecycle: from placement on market through use, repair, potentially re-use/repurpose and ultimately to end-of-life treatment. It consolidates product-related data in a standardized, machine-readable format and makes this data accessible to a set of actors depending on their role and authorization level. Essentially, a DPP links a physical product to a digital representation containing information about its materials, components, origin, carbon footprint, repairability, and other attributes relevant to sustainability and compliance.

The DPP serves regulators, businesses and consumers alike. For authorities such as Market Surveillance Authorities (MSA) and the European Commission (EC), it enables verification of product-related information and compliance with sustainability, safety, and information requirements defined under the Ecodesign for Sustainable Products Regulation (ESPR, Regulation EU 2024/1781) and, in the case of batteries, the Battery Regulation (Regulation EU 2023/1542). For businesses and consumers, it creates a shared information basis for more informed decisions across the value chain from procurement and use to maintenance and recycling. In this way, it can contribute to circular business models, improved traceability, and standardized data exchange across organizational and national boundaries.

However, the implementation of a DPP-compliant system poses significant challenges for organizations of all sizes. For large enterprises, the primary complexity lies in integrating DPP requirements into existing product data management systems, aligning internal data governance processes with external reporting obligations, and ensuring interoperability with the DPP systems of suppliers and downstream actors across the value chain as well as establishing effective collaboration across organizational units such as IT, production, after-sales, sustainability, and compliance. For small and medium-sized enterprises (SME), the challenge is more fundamental. SMEs frequently lack the dedicated technical resources, legal expertise, and financial capacity required to independently interpret the regulatory requirements, design a compliant system architecture, and maintain it as the framework evolves. The risk of misinterpretation of relevant regulations and standards or delayed implementation caused by the lack of IT personnel is considerably higher for smaller organizations.

The BatteryPass-Ready project seeks to address precisely this gap. The project aims to lower the barriers to compliant DPP implementation by providing a structured and accessible analysis of the data, technical and organizational requirements of digital product passport systems, supported by a testing and validation environment tailored for the needs of the digital battery passport (DBP).

2 SCOPE

This document derives the technical requirements of the battery passport from three main sources: the Battery Regulation, the ESPR and the standards and draft standards developed by CEN/CENELEC Joint Technical Committee 24 (JTC 24). These sources complement each other at different levels. The ESPR provides the general legal and architectural framework for Digital Product Passports across product sectors, while the Battery Regulation defines the first sector-specific implementation for batteries, including requirements for passport content, access, and lifecycle management. The JTC 24 standardization work further operationalizes these requirements through technical specifications for system architecture, data carriers, unique identifiers, APIs, interoperability, access rights management, and data integrity.

This document does not address the broader ESPR working plan, the rollout of DPP obligations to product sectors beyond batteries, or the delegated acts being developed in parallel under the ESPR framework. These aspects fall outside the scope of the BatteryPass-Ready project and remain under development at the time of writing. Where this document addresses data-related topics within the battery passport, it uses the Data Attribute Longlist Version 1.3 published in March 2026 developed by the BatteryPass-Ready consortium as the reference basis.

The failure and test scenarios developed in this document form the analytical foundation for the test environment being developed within the BatteryPass-Ready project. They are designed to capture the critical interaction points that a compliant DBP system must handle correctly.

The present document is structured as follows.

- Chapter 3 introduces the key regulatory, standardization, and technical documents that form the basis of the analysis.
- Chapter 4 establishes the technical and regulatory context of the DPP ecosystem. It covers the DPP system architecture, the actor model and access-rights framework, the relevant lifecycle and registry APIs, status codes, identifier types, and the authentication and authorization assumptions used for API access.
- Chapter 5 describes the methodology applied to translate regulatory and technical requirements into user stories, failure scenarios, and test scenarios.
- Chapter 6 presents the user stories for the business processes covered in this analysis, with a specific focus on the digital battery passport.
- Chapter 7 provides an outlook on planned next steps within the BatteryPass-Ready project.

3 KEY DOCUMENTS

This document draws on a set of key regulatory, standardization, and technical documents that together define the requirements, architecture, and implementation context for DPPs and, more specifically, the DBP. The documents described below represent the primary references used throughout this report.

The Ecodesign for Sustainable Products Regulation

The ESPR (Regulation EU 2024/1781) [1] was adopted on 13 June 2024 and entered into force on 18 July 2024. It replaces the previous Ecodesign Directive 2009/125/EC and significantly expands its scope, moving from energy-related products to practically all physical goods placed on the EU market, with limited exceptions for food, feed, and medicinal products. The Regulation establishes a framework for setting performance and information requirements, with the digital product passport serving as a key instrument for making product-related information available.

The ESPR does not define product-specific DPP content directly. Instead, it authorizes the European Commission to adopt delegated acts for individual product groups, specifying the required data attributes, data carrier, access conditions, and technical connection to the EC Registry. In this way, delegated acts translate the general ESPR framework into operational DPP requirements. The EC Registry is a central infrastructure component of the DPP system and is expected to be deployed by July 2026. Its technical and operational implementation is specified in the draft Commission Implementing Regulation laying down the implementation arrangements for the digital product passport registry (Draft Registry Act).

The Battery Regulation

The EU Battery Regulation (Regulation EU 2023/1542) [2] entered into force on 17 August 2023 and fully replaced the previous Battery Directive 2006/66/EC as of 18 August 2025. It directly mandates a DPP as a binding requirement for specific battery categories and thus complements the ESPR, which provides a general framework applicable across sectors. From 18 February 2027 specific battery categories including electric vehicle (EV) batteries, light means of transport (LMT) batteries, and rechargeable industrial batteries with a capacity greater than 2 kWh placed on the EU market must be electronically registered in the form of a DBP, accessible via QR code on the battery and in accompanying documents.

The DBP is not simply a static data record. It is a structured, role-based digital system that centralizes model-specific and battery-specific data including carbon footprint, recycled content, durability and performance parameters, as well as material and composition information. Access to this data is governed by defined roles, ensuring

that commercially sensitive information is protected while making relevant data available to authorized actors across the value chain.

The Battery Regulation also foresees a series of delegated and implementing acts that progressively define additional technical and operational details, including data formats, certification requirements for DPP service providers, and expanded content requirements. These acts are under active development at the time of writing. The regulation explicitly anticipates that further delegated acts will extend the scope of obligations.

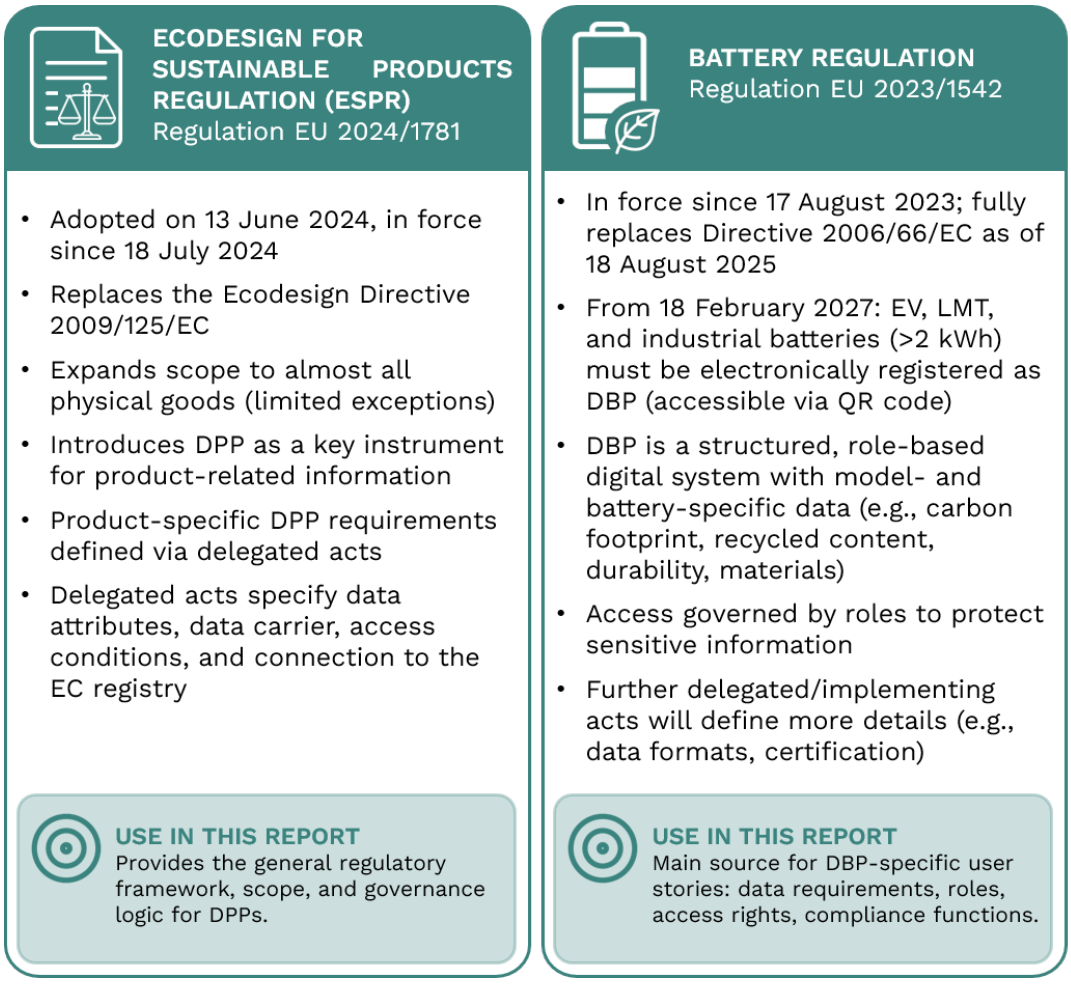


Figure 1: Summary of ESPR and Battery Regulation

The Draft Registry Act

The draft Commission Implementing Regulation on the digital product passport registry specifies how the EC Registry established under Article 13 of the ESPR is expected to operate. It sets out technical and operational rules for registering digital product passports, storing unique identifiers and relevant registration data, managing user access and identity verification, and enabling interaction between Economic Operators, value chain actors, competent authorities, customs authorities, and the Commission. It also defines key registry components, including the user interface, API, semantic repository, log system, and security and availability requirements. As the act is still in draft form, its provisions remain subject to change [3].

CEN/CENELEC JTC 24 Standardization

Following the standardization request addressed to the European standardization organizations CEN, CENELEC, and ETSI, CEN and CENELEC established JTC 24 to develop a harmonized family of standards for the DPP framework and system. These standards are intended to provide the technical basis for interoperable DPP systems across sectors, including their data architecture, operational requirements, access management, identifier schemes, and data integrity. The JTC 24 standards are used as technical reference documents for the analysis of DPP system requirements. They provide a structured basis for assessing how regulatory requirements can be reflected in the design and implementation of interoperable DPP systems. They also serve as a foundation for the creation of the test environment [4–11].

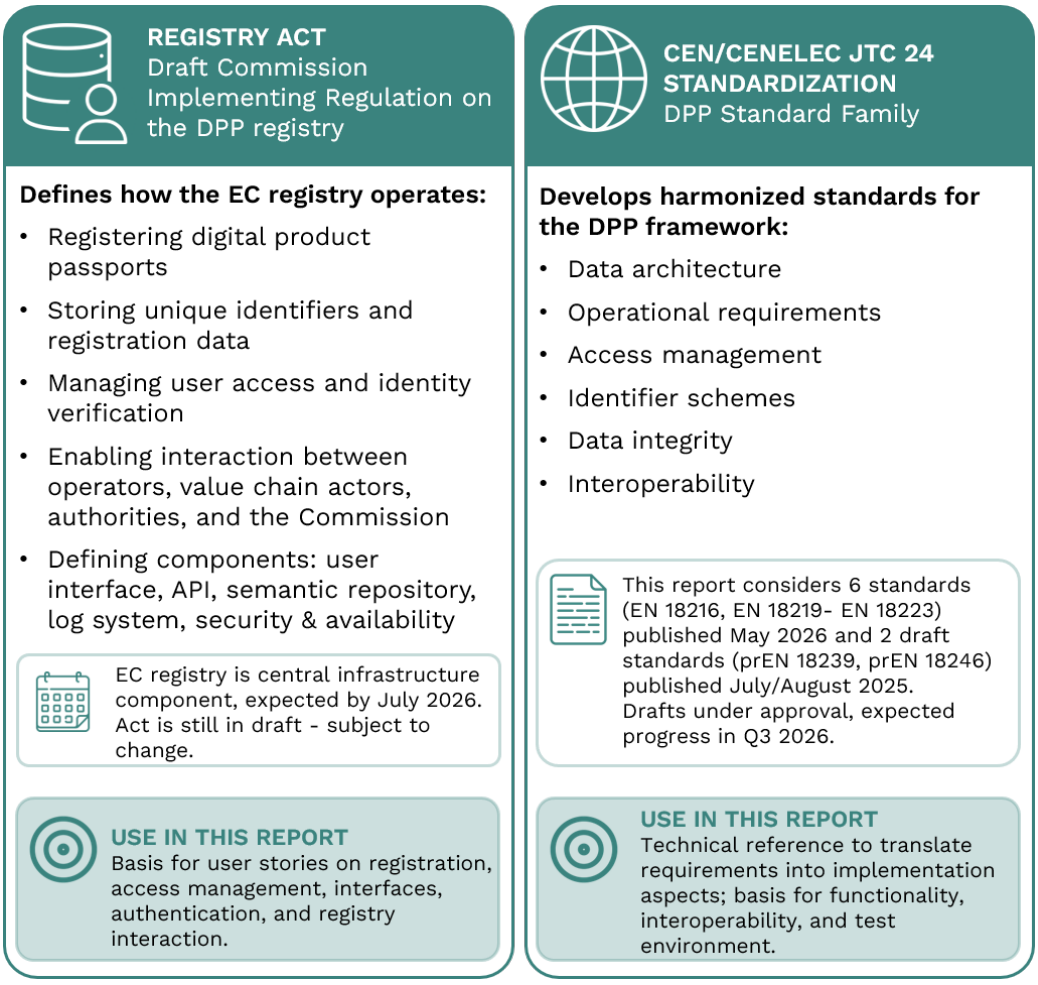


Figure 2: Summary of Draft Registry Act and JTC 24 Standards

Note: At the time of writing, the Commission Implementing Regulation on the digital product passport registry (Draft Registry Act) remains in draft form and has not yet been formally adopted. Its provisions are therefore still subject to change before publication. For the purposes of this document, the technical and operational rules set out in the current draft are taken as the authoritative basis for the analysis, and the processes described in the following user stories are built on the assumption that these draft provisions will apply substantially as currently specified.

4 OVERVIEW OF DPP ECOSYSTEM

4.1 DPP ARCHITECTURE

In the present context, the term *architecture* is used in a focused and limited sense. It refers to the structural organization of the DPP ecosystem, including the main components involved, the relationships between them, and the way information flows across the system. It does not address detailed software design, infrastructure sizing, hosting strategies, or vendor-specific implementation choices. These aspects remain within the responsibility of the organizations that develop and operate compliant solutions. The purpose of this chapter is to establish the structural perspective that provides the basis for the following sections.

For the purposes of this document, the analysis begins at the stage where battery related information is already available in a structured form and can be incorporated into the DPP ecosystem. The upstream processes through which data is generated, gathered, or consolidated by suppliers, manufacturers, logistics actors, or recyclers are not considered here. Instead, the focus lies on the ecosystem that stores, registers, makes available, protects, and exchanges passport information once that information can be used in a structured technical form.

DPP as a Combination of Data and System

DPPs can be understood as consisting of two closely connected but distinct elements: the passport data and the passport system.

The first element is the DPP data, i.e. the actual passport content such as product-related information, lifecycle-related information, and the identifiers needed to link the passport to a specific product and to the actors responsible for it. It is the content that must be stored, displayed, searched, and, where permitted, updated.

The second element is the system that makes this data operational. The system comprises the technical environment in which the passport is stored, registered, accessed, and protected. The distinction between data and system is important from an architectural perspective because it separates the passport content from the technical environment that manages and provides access to that content.

Core Architectural Components

The DPP system is, thus, not a single central application. It consists of several interacting technical components, each of which fulfils a specific function within the ecosystem.

The first core component is the operational DPP service. This service hosts the active version of the passport and provides the technical interface through which passport

data can be accessed, retrieved, and, where permitted, updated. The operational DPP service is the main access point for the active passport during its lifecycle.

The second core component is the backup service. This component stores a backup copy of the passport. Its function is not to manage the active lifecycle of the DPP, but to support continuity, resilience, and long-term availability in cases where the primary operational service is no longer available.

The third core component is the EC Registry. The EC Registry does not store the full passport content. Instead, it acts as a central reference and indexing layer. It contains the identifying information needed to register a DPP and make it discoverable within the wider DPP ecosystem. In this way, the registry supports identification and discoverability, while the actual passport content remains outside the registry.

The fourth core component is the communication and interface layer. This includes the APIs through which the different components interact with each other. APIs enable structured technical requests and responses, for example to register a passport, retrieve a passport, access selected data elements, or update information where this is allowed. The architecture must also support physical or digital access points through which users can reach the passport information via QR codes. Such access points are particularly relevant for public access to non-sensitive passport information

The fifth core component is the access-control and security layer. This layer governs which data can be accessed, by whom, and under which conditions. It is therefore closely connected to authentication, authorization, secure data exchange, and the distinction between public and controlled passport information.

Communication Logic and APIs

The components of the DPP ecosystem interact through APIs. From an architectural perspective, this means that communication between system components takes place through structured technical requests and responses rather than through informal or manual exchange.

These interactions are largely based on identifiers. Requests typically refer to a DPP identifier, a product identifier, or a path to a particular data element. The response may then contain the passport, a selected part of the passport data, a list of relevant identifiers, or a result object indicating the outcome of the request.

In addition to API-based communication, the EC Registry is expected to provide a web-based user interface. While APIs support automated system-to-system interaction, the web interface may support direct human interaction with the EC Registry, for example for registration-related, administrative, or consultation functions, depending on the final implementation of the registry.

Architectural Implications

The architecture of the DPP ecosystem can therefore be summarized as a decentralized system with a centralized reference layer. The EC Registry supports registration and discoverability, while the operational passport content is hosted outside the

registry. Backup services support resilience and long-term availability. APIs connect the components and enable structured interaction across the ecosystem. Access-control and security mechanisms ensure that public and controlled information are made available only through appropriate access paths.

4.2 ACTORS AND THEIR ROLES IN DPP SYSTEMS

4.2.1 ACTORS IN DPP SYSTEMS

At the regulatory level, the Battery Regulation assigns different access rights to different actors. It distinguishes between information available to the public, information accessible only to authorities and notified bodies, and information available only to actors with a legitimate interest, such as those involved in repair, remanufacturing, second-life use, or recycling.

For this document, however, a purely legal list of stakeholders is not sufficient. Since the focus is on technical specifications, the actor model must also reflect how participants interact with DPP services and access-control mechanisms. Therefore, this document follows the more implementation-oriented approach of the JTC 24 drafts.

Accordingly, seven actor groups are used: custodians, Backup Service Providers (BSP), the EC Registry, authorities, regulated base roles, other third parties, and public users. Custodians include Economic Operators and, where applicable, Main Service Providers (MSP). The EC Registry is treated as a special actor-like interaction point: although it is architecturally a central component, DPP systems interact with it through defined registry functions.

This classification remains aligned with the regulatory framework while providing a structure better suited for technical design, access-rights modelling, and API interactions.

Custodians

Custodian refers to the actor that operates or controls the active DPP interface for a product. The custodian is an Economic Operators or a Main Service Provider [8].

The Economic Operator is the primary responsible party in the DPP lifecycle. Depending on the legal and product-specific context, this role may include manufacturers, importers, distributors, dealers, or other operators responsible for placing a product on the market. From a technical perspective, the Economic Operator is important because it is responsible for ensuring that the DPP exists, for storing and making required data available, and for managing access to the passport in accordance with the applicable rules.

In many practical contexts, Economic Operators may not operate DPP infrastructure directly. Instead, they may authorize a Main Service Provider to host the passport, provide the required interfaces, manage the technical availability of the DPP, and enforce access rights. In such cases, the Main Service Provider acts as a custodian on behalf of the Economic Operator. However, this operational delegation does not transfer the overall responsibility for the DPP away from the Economic Operator.

At the same time, the role of the Main Service Provider remains limited to the provision of the agreed DPP-related services. This limitation is reflected both in the ESPR and in the Battery Regulation. For the battery passport specifically, Article 78(d) of the Battery Regulation states that, where battery passport data is stored or otherwise processed by operators authorized to act on behalf of the Economic Operator

responsible for the obligations under Article 77(4) or Article 77(7), those operators may not sell, re-use, or process such data, in whole or in part, beyond what is necessary for providing the relevant storing or processing services. The ESPR follows the same logic by limiting service providers' use of DPP data to what is necessary for the agreed service, unless this has been expressly agreed with the relevant Economic Operator. This limitation is important for the actor model because it clarifies that service providers may operate technical functions on behalf of the Economic Operator, but do not obtain independent rights over the DPP data.

Backup Service Providers

A second group is formed by backup digital product passport service providers. Their role is to support the availability, continuity, and recoverability of digital product passport data over the passport lifetime. They are not responsible for the day-to-day editing of passport data, but for hosting a backup copy of the passport and relevant historical versions.

The relevance of this role is also reflected in the DPP registration framework. Upon registration, the Commission verifies, where relevant, the link to the backup hosted by a digital product passport service provider. In addition, the DPP Registry Entry foresees metadata such as the identifier of the backup operator.

Although it is not clearly described that battery passports require both backup and archiving in all cases, this work assumes based on Battery Regulation Art. 78 (e) that such mechanisms are necessary to prevent loss of access to passport data if the Economic Operator's system fails or if the Economic Operator leaves the market. Therefore, Backup Service Providers are considered essential for resilience and long-term accessibility of battery passport information.

EC Registry

The third actor is the EC Registry. It represents a special case in the actor model because it is not an actor in the same sense as an Economic Operator, service provider, authority, or public user. From an architectural perspective, the EC Registry is a central technical component of the DPP ecosystem. It does not store the full content of digital product passports but acts as a reference and indexing layer through which DPPs are registered and made discoverable.

Nevertheless, the EC Registry is included in the actor model as an actor-like interaction point. This is useful for technical modelling because compliant DPP solutions must communicate with it through defined registry functions, for example when a new passport is registered or when identifying information is used for discovery purposes.

Its role is therefore limited but central. The EC Registry supports registration, identification, and discoverability. It does not manage the active passport content and does not replace the operational DPP service hosted by the responsible custodian.

Authorities

The fourth actor group are authorities, which interact with the DPP ecosystem mainly for supervisory, enforcement, and compliance-related purposes. This group may include market surveillance authorities, customs authorities, notified bodies, and the European Commission.

From a technical perspective, authorities are relevant because they require trusted and controlled access to DPP information. Their access may include data that is not publicly available. Therefore, the system must support secure identification, authentication, authorization, and reliable retrieval of the information required for their official tasks.

Regulated Base Roles

The fifth actor group consists of regulated base roles. This category includes actors whose access rights are linked to formally recognized roles in the battery lifecycle, such as recyclers or refurbishers. These actors require access to specific controlled, non-public DPP data in order to perform their respective activities.

The distinguishing feature of this group is that their role and related access rights are defined through the applicable legal framework and related implementing acts. Access to the relevant controlled data may therefore be based on the verified role of the actor, rather than on separate approval by each individual Economic Operator, provided that the applicable authentication or certification mechanism allows this role to be reliably verified.

Third Parties

The sixth actor group is formed by other third parties. This category covers actors whose access rights are not directly defined through a regulated base role. These actors are also not the primary custodian of the DPP and do not operate the active passport interface.

Their access is typically limited, contextual, and dependent on permissions granted by the responsible Economic Operator or custodian. This may include business customers, service partners, or other parties that need access to selected DPP information for a specific purpose. Unlike regulated base roles, these actors do not automatically receive access based on a generally recognized legal role. Instead, their access must be authorized according to the relevant activity, contractual relationship, or permission model.

Public Users

The final actor group is formed by public users. This category covers all users who access publicly available DPP information without authentication. Although public users are not always identified as a separate actor group across all standards, they are relevant for technical implementation because every DPP system must provide access to public passport data.

According to prEN 18239, access to public data must be possible for both European Union (EU) and non-EU actors without additional authentication. The Battery Regulation follows the same logic by distinguishing information intended for the public from information subject to restricted access. For this reason, unauthenticated readers of public passport information are treated as a separate actor group in this document. Public users may access DPP data, for example, by scanning a QR code placed on the product. Their access is limited to non-sensitive information classified as public under the applicable delegated acts. They do not receive access to controlled, confidential, or role-specific data.

Taken together, the actor classification used in this document provides a practical bridge between regulation and technical implementation. It preserves the meaning of the legal framework while reorganizing the DPP ecosystem into actor groups that are suitable for system architecture, access-rights modelling, and API planning. Figure 3 provides an overview of the main actor categories and their key characteristics as adopted in this document.

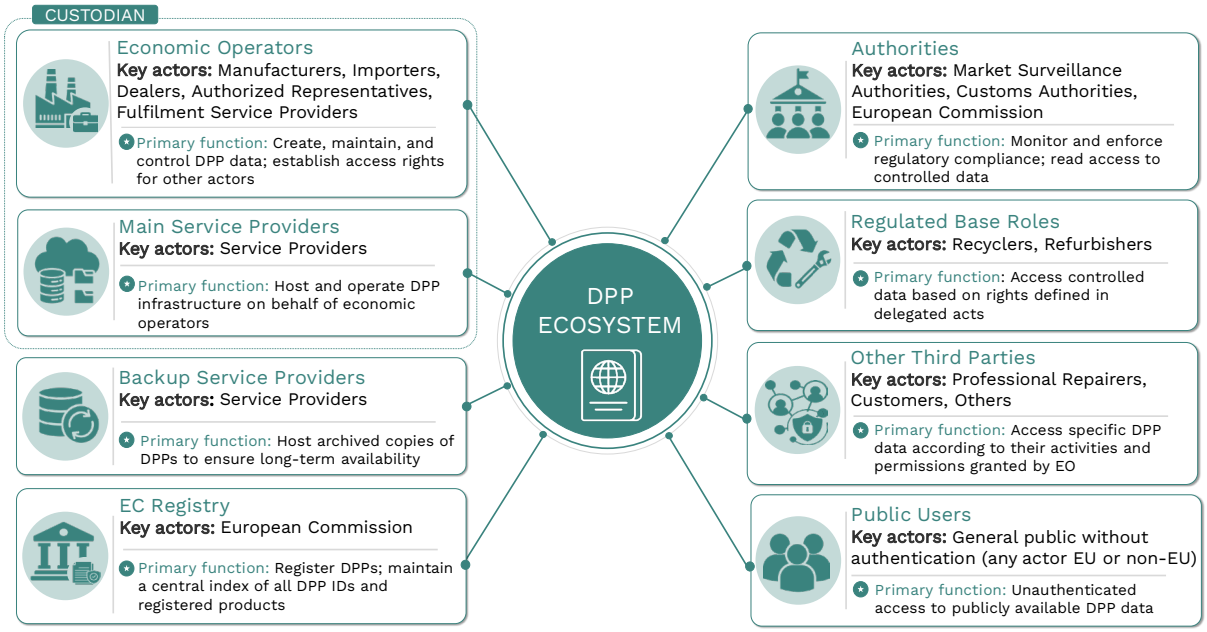


Figure 3: Classification of Actors in the DPP Ecosystem

4.2.2 ACCESS RIGHTS MANAGEMENT SYSTEM

The actor categories introduced in Chapter 4.2.1 serve a functional purpose within the DPP system. They do not simply describe the participants in the ecosystem but determine how the system must handle access to information in practice. Each category is linked to a defined set of permissions that specifies which data may be accessed, whether the actor is limited to reading information or may also modify it, and under which conditions a technical request is accepted or rejected by the system. Access rights management therefore forms an essential part of DPP operation.

Its importance can be understood on three levels. First, it is a legal requirement for the battery passport, since the Battery Regulation, within the broader framework of the ESPR, requires access to DPP information to be limited according to defined actor categories. Second, it is a governance tool, because the allocation of access

rights to specific battery-related data must be determined through the applicable implementing acts. Third, it is a technical condition for system functionality, because access rights determine which operations an actor may perform within the DPP system and which API requests the system is allowed to process.

The central technical reference in this area is prEN 18239, which defines requirements for access rights management, access control, information security, and the protection of business confidentiality within the DPP ecosystem. For the battery passport, these requirements must be read together with Article 77(9) of the Battery Regulation, which requires the Commission to specify which persons are considered to have a legitimate interest, which information they may access, and to what extent they may download, share, publish, or re-use it. Access rights management therefore provides the technical mechanism for implementing these regulatory access rules in the battery passport system. In this context, the forthcoming implementing act on Access Rights to Certain Parts of the Battery Passport, for which Commission adoption is planned for Q4 2026, will further specify the regulatory framework and is therefore expected to play an important role in shaping the practical implementation of access rights within the battery passport.

Public Data and Controlled Data

A core distinction in access rights management is the distinction between public data and controlled data. Public data refers to information that can be accessed without authentication. According to prEN 18239, such access must be possible for both EU and non-EU actors without additional authentication. This means that members of the public, for example users scanning a QR code on a product, may retrieve this information directly and without authentication. Public data therefore constitutes the openly accessible, read-only part of the passport.

Controlled data, by contrast, may only be accessed by actors whose identity has been authenticated and whose authorization has been verified. This category typically includes information that is not intended for unrestricted public access, for example because it is commercially sensitive, operationally relevant, personal in nature, or could be used to derive sensitive business or usage patterns.

Access rights for such data are not assigned only at the level of the passport as a whole. Instead, they must be defined for each controlled data element individually and expressed through controlled vocabularies. As a result, different actor categories may receive access to different parts of the battery passport depending on their role, their legal access basis, and the applicable implementing acts.

Figure 4 summarizes the general access logic used in this document. It reflects the distinction between public and controlled data and shows how actor categories are linked to different levels of access within the battery passport system.

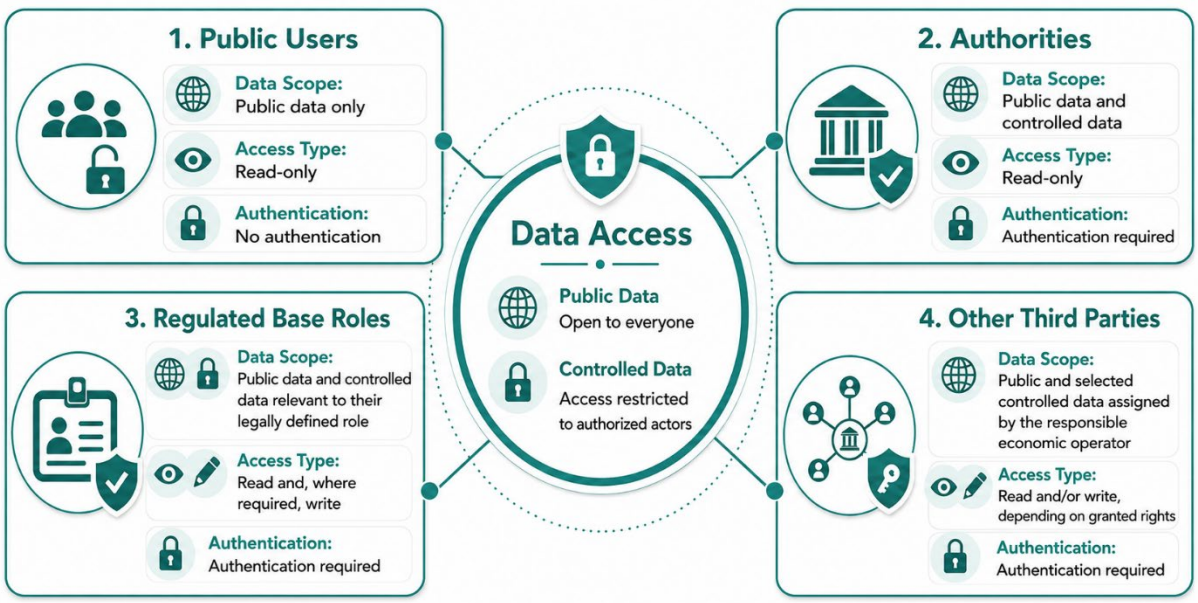


Figure 4: Actor-Based Data Access Rules in the DPP Ecosystem

Note: It should be noted that this figure summarizes access at actor-category level. In practice, prEN 18239 requires access rights to be defined for each controlled data element. This applies to actors requesting authorized access to controlled data, meaning that the exact set of readable or writable fields may differ even within the same actor category.

4.3 PRODUCT PASSPORT LIFECYCLE APIS

An API is a defined set of methods that enables one software component to interact with another. It allows different systems to exchange information in a structured, predictable, and machine-readable form without requiring either system to understand the internal design of the other. Instead of relying on a person to read information in one system and manually transfer it into another, an API allows applications to request, send, and process data automatically according to a defined protocol.

In digital environments, APIs are the standard means of communication between system components. This is especially important when those components are developed by different organizations, operated by different actors, or hosted in different technical environments. A system sends a request with defined inputs, the API performs the required function, and the result is returned in a structured format.

Standard EN 18222 structures this API landscape into three functional areas: the Life Cycle API, the Fine Granular Life Cycle API, and the EC Registry API. Together, these interfaces cover the main technical operations required during the lifecycle of a DPP. They support the creation and retrieval of passports, their registration in the EC Registry, and the reading, updating, or deletion of specific content where more detailed control is needed.

Life Cycle API

The Life Cycle API forms the central functional layer of the DPP system. It is provided by custodians, meaning Economic Operators or Main Service Providers acting on their behalf, and, where relevant, by archiving services operated by Backup Service Providers. The methods in this group support the creation, retrieval, update, and deletion of complete DPP records (see Table 1).

Table 1: Overview of the Life Cycle APIs according to EN 18222

Method Name	HTTP Method	Main Input	Main Output	Function Description	Provided by
ReadDPPById	GET	DPP ID	Status code, DPP	Returns the DPP with the specified DPP ID	EO, MSP, BSP (mandatory)
ReadDPPByProductId	GET	Product ID	Status code, DPP	Returns the current active DPP with the specified Product ID	EO, MSP, BSP (mandatory)
ReadDPPIdsByProductIds	POST	Set of product IDs, optional limit and cursor	Status code, set of DPP IDs	Returns a list of DPP IDs matching Product IDs	EO, MSP, BSP (mandatory)

Method Name	HTTP Method	Main Input	Main Output	Function Description	Provided by
ReadDPPVersioByIdAndDate	GET	Product ID, timestamp for which the DPP is requested	Status code, DPP	Returns a DPP version based on its DPP ID and given date	MSP, BSP (recommended) EO (optional)
CreateDPP	POST	Complete DPP payload	Status code, DPP ID	Creates a new DPP	MSP, BSP (recommended)
UpdateDPPById	PATCH	DPP ID, partial DPP payload	Status code, updated DPP	Partially updates an existing DPP	EO, MSP (mandatory)
DeleteDPPById	DELETE	DPP ID	Status code	Deletes DPP with the specified DPP ID	MSP, BSP (recommended)

From the perspective of access rights, the READ methods in this API group are available to actors who hold the necessary permissions for the requested data. By contrast, CreateDPP and UpdateDPPById are write operations and may only be used by actors with write authorization. This primarily includes custodians and, where explicitly permitted, authorized third parties such as actors fulfilling regulated base roles. DeleteDPPById is likewise subject to strict access control and is expected to be used only under defined conditions at the end of the passport's operational life by the responsible Economic Operator.

Note: It is assumed that the DeleteDPPByID API is only available to the responsible Economic Operator. Therefore, only the Economic Operator may initiate or authorize the deletion of a DPP. This assumption is based on the ESPR, which assigns key DPP responsibilities to the Economic Operator.

Fine Granular Life Cycle API

The Fine Granular Life Cycle API complements the main lifecycle interface by enabling interaction with individual data elements within a DPP rather than with the passport as a whole. This is particularly important because, as discussed in Chapter 4.2.2, access rights may differ at the level of individual controlled data elements. Where such differentiated permissions apply, fine-granular operations allow authorized actors to read or update only the specific part of the passport that falls within their scope of access. According to EN 18222, these operations may be provided by custodians or BSP (see Table 2).

Table 2: Overview of Fine Granular Life Cycle API according to EN 18222

Method Name	HTTP Method	Main Input	Main Output	Function Description
ReadDataElement	GET	DPP ID, path to the specific data element	Status code, DataElement	Returns a single data element
UpdateDataElement	PATCH	DPP ID, path to the specific data element, content of data that needs to be updated	Status code, updated element	Updates a specific data element as well as amend or remove data element information

This interface is especially relevant in lifecycle situations where only selected data fields need to be changed. By addressing individual elements, it supports efficient updates while allowing the system to enforce field-level access control. In this way, it directly reflects the requirement in prEN 18239 that access rights for controlled data be defined at the level of individual data elements.

EC Registry API

The EC Registry API has a more limited functional scope, but it plays an essential structural role within the DPP architecture. It provides the interface through which a newly created DPP is registered in the EC Registry. For this purpose, EN 18222 defines the method RegisterProductDPP. This method is provided by the EC Registry server and returns a Registration ID to the requesting client Economic Operator (see Table 3).

Table 3: Registry API according to EN 18222

Method Name	HTTP Method	Main Input	Main Output	Function Description
RegisterProductDPP	POST	dppRegistryEntry: • Product ID • DPP ID • Economic Operator ID • Economic Operator ID of the Backup provider • DPP API end point	Status code, Registration ID	Registers a new DPP in the EC Registry

The request is initiated by the custodian Economic Operator or by the Main Service Provider acting on its behalf. The input is provided as a dppRegistryEntry and is limited to the information required for registration and discoverability. The request therefore does not contain the full passport content. The Registration ID returned by the server can subsequently support traceability and enforcement activities by competent authorities and provide proof that the product has been registered.

Status Codes

Every API method defined in EN 18222 returns a status code as part of the response. This code informs the requesting system whether the operation was successful or failed. If the request was not completed successfully, the response usually also contains a result object with further details that help the requesting system understand the reason for the failure and respond accordingly.

EN 18222 specifies a standardized set of status codes that represent the main possible outcomes of an API call. The following table distinguishes between successful outcomes, client-side errors, and server-side errors, and outlines the meaning of each code in the context of DPP API interactions (see Table 4).

Table 4: Overview of Status Codes according to EN 18222

Status Code	Category	Meaning	HTTP Status Code
Success	Success	The operation was completed successfully	200
SuccessCreated	Success	A new resource was successfully created	201
SuccessAccepted	Success	The reception of the request was successful	202
SuccessNoContent	Success	Success with explicitly no content in the payload	204
ClientErrorBadRequest	Client error	The request was malformed or otherwise invalid	400
ClientNotAuthorized	Client error	Credentials are missing or incorrect	401
ClientForbidden	Client error	Authorization has been explicitly refused	403
ClientMethodNotAllowed	Client error	Method request is not allowed	405
ClientErrorResourceNotFound	Client error	The requested DPP or data element does not exist	404
ClientResourceConflict	Client error	A conflict occurred, for example because the resource already exists	409
ServerInternalError	Server error	An unexpected internal error occurred	500
ServerNotImplemented	Server error	The server does not support the functionality to fulfil the request	501
ServerErrorBadGateway	Server error	A gateway-related error occurred	502

From a technical perspective, these standardized response codes are important for two reasons. First, they allow requesting systems to process API responses in a consistent and automated way across different DPP service providers. Second, they support traceability and auditability by making access attempts and failed operations easier to document and evaluate in line with the requirements of prEN 18239.

4.4 IDENTIFIERS

The DPP framework relies on a structured system of unique identifiers that link physical products to their digital representations. Identifiers underpin access control, discoverability, enforcement, and trust across the entire DPP ecosystem.

Five identifiers are established by the JTC 24 standards:

Unique Battery Identifier (Battery ID) / Unique Product Identifier (Product ID)

The Battery ID / Product ID is a unique string of characters that identifies a product and enables the link to its digital product passport. The product must be connected through a data carrier to a persistent Uniform Resource Identifier (URI). In the battery passport context, this access is provided via a QR code linked to the Product ID assigned to the battery by the Economic Operator. While the Product ID may generally be issued at three levels of granularity, namely model, batch, or individual item, the battery passport requires identification at the individual item level.

Unique Battery Passport Identifier (DBP ID) / Unique Digital Product Passport Identifier (DPP ID)

The DBP ID / DPP ID identifies a specific DPP instance. According to EN 18223, it is a mandatory attribute of each DPP object and consists of a globally unique string of characters that should be structured as a Uniform Resource Identifier (URI) or Uniform Resource Locator (URL).

Unique Economic Operator Identifier (Economic Operator ID)

The Economic Operator ID is a unique string value that links the DPP to the Economic Operator responsible for it. Where a Backup Service Provider is involved, the corresponding identifier is used to link the DPP to the actor responsible for the backup function.

Unique Facility Identifier (Facility ID)

The Facility ID is a unique string of characters for the identification of location or building involved in a product's value chain or used by actors involved in a product's value chain.

Unique Registration Identifier (Registration ID)

The Registration ID is issued once an Economic Operator uploads the necessary Identifiers needed for registering a DPP with the EC Registry. The EC Registry returns a unique registration identifier associated with the uploaded data. This identifier serves as proof of registration.

All identifier types must comply with internationally recognized standards, in particular the ISO/IEC 15459 series, which governs unique identification systems for products, groupings, and supply chain entities [7].

From a technical perspective, identifiers are essential for the DPP APIs described in Section 4.3. They are used to address specific resources in requests and to confirm created or retrieved resources in responses. A clear understanding of the relevant identifier types, their scope, and the applicable standards is therefore necessary for implementing and testing the interaction sequences described in the following user stories (see Table 5).

Table 5: Identifiers

Identifier Type	Definition	Role in DPP System
Unique Product Identifier	Unique string identifying a product and enabling a web link to its DPP	Links physical product to digital passport via data carrier; required at model, batch, or item level
Unique Digital Product Passport Identifier	Unique string identifying a specific DPP instance.	Uniquely identifies a DPP instance within the backend system; used as the primary input for API operations including retrieval, update, and deletion of passport records.
Unique Economic Operator Identifier	Unique string identifying an actor in a product's value chain	Required in DPP for manufacturers and other actors; forms basis for access rights assignment and authentication
Unique Facility Identifier	Unique string identifying locations or buildings in a product's value chain	Included in DPP to trace manufacturing and handling facilities
Unique Registration Identifier	Identifier automatically issued by the EC Registry upon DPP registration	Returned to Economic Operator after upload; required for customs clearance; enables enforcement access

4.5 AUTHENTICATION AND AUTHORIZATION FOR API ACCESS

The previous chapters have shown that the battery passport system distinguishes between public and controlled data. Public data can be accessed without additional authentication, whereas controlled data may only be accessed by actors whose identity and access rights have been verified. For protected API interactions, the system must therefore answer two different questions: who is making the request, and whether this actor is allowed to perform the requested action.

In this context, authentication and authorization must be clearly distinguished. Authentication refers to the process of verifying the identity of the requesting actor, for example an Economic Operator, authority, recycler, repairer, or service provider. Authorization refers to the subsequent decision on what this authenticated actor is allowed to do. It determines whether the actor may access a specific data element, whether access is limited to reading, or whether the actor may also create, update, or modify information. Authentication therefore establishes who the actor is, while authorization determines what this actor may do within the battery passport system.

At the time of writing, the regulatory and standardization framework does not define one single mandatory authentication method for all DPP interactions. For this reason, the present document does not treat a specific mechanism as a final normative requirement. Instead, it defines an implementation assumption for the user stories developed in the following chapters. This assumption is based on the origin of the access right and on the distinction between regulated access and operator-managed access.

Access rights may arise in different ways. Some actors, such as authorities or actors fulfilling regulated lifecycle roles, derive their access rights from legal provisions, implementing acts, or official functions. Other actors, such as service partners or selected third parties, may receive access from the responsible Economic Operator. This distinction is relevant because the system must not only verify the identity of the requester but also determine whether the requester is acting in a role that permits access to the requested data.

Regulated Access

For authorities and actors whose access rights are based on regulated base roles, authentication must go beyond a simple local user account. The system must be able to verify that an actor is acting in a recognized role or official capacity. For example, a recycler requesting controlled end-of-life information or an authority requesting restricted compliance information must be identifiable through a trustworthy mechanism that supports role-based or attribute-based verification.

In this context, eIDAS-based mechanisms can be considered a plausible implementation path for trusted identification and regulated access control. Regulation (EU) 2024/1183, adopted on 11 April 2024 and in force since 20 May 2024, amended the original eIDAS Regulation (EU) No 910/2014, to establish the European Digital Identity Framework. This reform is commonly referred to as eIDAS 2.0. One of its central elements is the European Digital Identity Wallet, which enables natural and legal

persons to use trusted electronic identification and electronic attestations of attributes [12].

These mechanisms can complement identity verification by allowing users to present verified attributes, such as professional qualifications, public permits, powers of representation, mandates, or the authority to act on behalf of a legal person. For DPP scenarios, this is particularly relevant where access to specific passport data or functions depends on a verified stakeholder role, regulated status, or official mandate. However, the use of eIDAS-based mechanisms would need to be aligned with the applicable DPP access rights, sector-specific requirements, and the availability of authentic sources for the relevant attributes.

Economic Operator-Managed Access

Not all access to controlled DPP data is expected to arise directly from regulated roles. In some cases, the responsible Economic Operator or a Main Service Provider acting on its behalf may grant access to selected third parties, such as service partners, suppliers, repair partners, or other actors with a specific operational need.

For these operator-managed access scenarios, token-based mechanisms such as OAuth 2.0 and OpenID Connect are treated in this document as a suitable implementation assumption. They allow access to be granted through tokens limited by scopes, expiry periods, and permissions. This supports the principle of least privilege, since a third party can be authorized only for the specific operations and data fields required for its task.

In this model, OpenID Connect supports the authentication of the user or entity behind the request, while OAuth 2.0 supports the authorization of specific API operations. The Economic Operator or its Main Service Provider can therefore authenticate a third party, issue appropriately scoped tokens, and enforce read-only or read-write permissions at API level.

Assumptions

The following user stories are based on a differentiated authentication model. eIDAS-based mechanisms are assumed for authorities and actors whose access rights depend on regulated base roles and official functions. OAuth 2.0 and OpenID Connect are assumed for third parties whose access is granted and managed by the responsible Economic Operator or by the Main Service Provider acting on its behalf.

These assumptions are used to make the technical interaction scenarios testable and internally consistent. They should not be understood as a final prescription of the only permissible authentication mechanisms. If future standards, delegated acts, or implementing acts define more specific authentication requirements, the assumptions used in this document must be updated accordingly.

5 METHODOLOGY

Approach

The methodology applied in this document follows a structured, process-oriented approach to translating regulatory and technical requirements into testable system interactions. Starting from the relevant obligations defined in the Battery Regulation, the ESPR, the Draft Registry Act, and the JTC 24 standards, the analysis identifies the core business processes that a DPP system must support and breaks them down into their constituent technical steps. The result is a set of user stories that describe, in precise but accessible terms, what the system must do, who is involved, and how interactions between actors and system components are expected to unfold. Each user story also defines what can go wrong and how correct behavior can be verified, making the document directly usable as the basis for a test environment.

Business Processes

A business process, in the context of this document, refers to a distinct operational activity that an actor or group of actors must carry out within the battery passport lifecycle. Each business process has a defined purpose, a defined set of participants, and a defined outcome. Business processes form the top-level organizational unit of the user story structure in Chapter 6. Each chapter section corresponds to one business process, which may be further divided into individual user stories where multiple distinct interaction paths exist.

User Stories

A user story describes a specific interaction scenario within a business process from a technical perspective. In this document, user stories describe the overall interaction flow between all involved actors and system components, rather than the actions of one individual user. Each user story is self-contained and can be read independently, though it may reference modules or concepts defined elsewhere in the document.

Each user story in Chapter 6 follows a consistent structure composed of the following elements.

Overview: Each user story opens with a concise summary of its purpose, and the scope of the interaction being described. It also states any assumptions that apply to the scenario.

Actors and Roles: Each user story defines the actors involved in the user story and their specific role.

Ordinary Sequence: The ordinary sequence describes the expected technical flow of interactions between actors and system components when all preconditions are met and no errors occur. It is written as a step-by-step narrative that follows the module chain defined for the user story.

Failure scenarios: A failure scenario describes a specific deviation from the ordinary sequence that a compliant DPP system must be able to prevent or handle correctly. Failure scenarios are derived directly from the ordinary sequence of a user story by identifying points at which invalid inputs, missing credentials, system errors, or conflicting states could cause the interaction to break down.

Test scenarios: A test scenario defines the conditions under which the system's correct behavior must be verified. Each test scenario specifies a test objective, the preconditions that must be in place, and the expected outcome that constitutes a passing result. Test scenarios address both the normal flow and the failure cases, ensuring that the system handles both correctly.

Modules

To ensure consistency across all user stories, the ordinary sequences are composed from a defined set of reusable modules. Each module represents a self-contained technical interaction that can appear in multiple user stories and in different positions within a sequence. The seven modules used in this document are:

Authentication and Authorization: Covers the verification of an actor's identity and the confirmation of their access rights before any operation is processed.

Registration: Covers the submission of a set of IDs to the EC Registry and the receipt of the Registration ID with the purpose of registering a DPP.

Create DPP: Transfer DPP data from the EO to the MSP to create a DPP that is subsequently hosted by the MSP.

Read: Covers the retrieval of DPP data, either publicly without authentication (QR Code Read) or in a controlled manner (Controlled Data Read) following authentication and authorization.

Update: Covers the modification of one or more data elements within an existing DPP.

Backup: Covers the creation or update of a backup copy at the Backup Service Provider following a registration or update operation.

Deletion: Covers the removal of an active DPP from public hosting, including soft deletion at the Backup Service Provider and notification of the EC Registry.

6 USER STORIES

This chapter applies the methodology described above to the core business processes of the digital battery passport. While the preceding chapters established the regulatory, architectural, and technical context of the DPP ecosystem, the following user stories translate this context into concrete interaction scenarios for the DBP. In total, this chapter presents eight user stories.

Figure 5 provides an overview of the user stories presented in this chapter.

User Story	Module Order
Economic Operator places the Battery on the Market	Authentication & Autorization by BSP → Backup → Authentication & Autorization by EC Registry → Registration
Main Service Provider places the Battery on the Market	Authentication & Autorization by MSP → Create DBP → Authentication & Autorization by BSP → Backup → Authentication & Autorization by EC Registry → Registration
Reading Public Battery Passport Data	Read QR Code
Reading Controlled Battery Passport Data	Authentication & Autorization by EO / MSP → Read Conntrolled Data
Search for Battery Passports	Authentication & Autorization by EO / MSP → Read Controlled Data
Updating of Battery Passport Data	Authentication & Autorization by EO / MSP → Update → Authentication & Autorization by BSP → Backup
Deletion of Battery Passport	Authentication & Autorization by BSP→ Deletion→ Authentication & Autorization by EC Registry → Registry Notification
End-of-Life and Circularity Events	Read QR Code → Authentication & Autorization by EO / MSP → Read Controlled Data → Update → Read Controlled Data

Figure 5: List of User Stories

Note: The failure and test scenarios presented in this document provide a broad overview of potential failure conditions and critical interaction points relevant to the digital battery passport. The scenarios constitute a superset of what the test environment developed within the BatteryPass-Ready project will ultimately implement. The test environment will cover a prioritized and technically reproducible subset of these scenarios. Scenarios that depend on physical conditions, on external systems, or on behaviours not yet fully specified by the still-evolving delegated acts may not be reflected in the implemented tests. The scenarios should therefore be read as an analytical reference for robust DBP system design, not as a one-to-one specification of the test environment's coverage.

6.1 REGISTER BATTERY AND PLACE IT ON THE MARKET

The placement of a battery on the market is defined as the overarching business process. Within this business process, three distinct user stories are considered. The first user story considers the initial onboarding on the registry website and creation of a profile. The second user story describes the scenario in which the Economic Operator carries out all required activities independently. The third user story covers the scenario in which the Main Service Provider acts on behalf of the Economic Operator.

Before a battery can be placed on the market, it must be registered in the EC Registry. This step represents the starting point of the battery passport lifetime. Within the scope of this project, no assessment is made regarding how the required data has been collected. Responsibility for the availability, correctness, and completeness of the data lies solely with the Economic Operator. It is therefore assumed that all necessary information is already available, that the digital battery passport has been fully prepared, and is ready for registration.

Note: The user stories in this chapter are based on the current draft of the Registry Act and the registry-related assumptions available at the time of writing. Since the final implementation of the EC Registry is expected to become available in mid-July 2026, some process steps may need to be reviewed and refined accordingly.

6.1.1 REGISTRY ONBOARDING AND ACCESS RIGHTS MANAGEMENT

Overview

This user story describes the one-time onboarding process that must be completed before the first battery passport registration. It includes authentication and authorization of the EO and the MSP in the EC Registry, verification of their identities, and setup of their registry user profiles. After successful onboarding, the EO can either register battery passports itself or grant the MSP the access rights required to perform registration on its behalf. After this process has been completed, the MSP is authorized to act on behalf of the EO in subsequent battery passport registration processes[3].

Actors and Roles

Actor	Role	Responsibilities	Authentication Protocol provided by
Economic Operator	Responsible Actor / Access Rights Owner	Authenticates with the EC Registry, verifies its identity, sets up and maintains its user profile, registers battery passports directly, or grants access rights to an MSP acting on its behalf.	
Main Service Provider	Service Provider	Authenticates with the EC Registry, verifies its identity, sets up and maintains its user profile, and receives access	

Actor	Role	Responsibilities	Authentication Protocol provided by
		rights from the EO to act on its behalf during registration.	
EC Registry	Registration Authority	Controls access to registry services, verifies the identity of the EO and MSP, supports user profile setup, and records the access rights granted by the EO to the MSP.	eIDAS

Ordinary Sequence

The ordinary sequence starts with the authentication of the EO or MSP by the EC Registry. Actors authenticate with the EC Registry using eIDAS to establish its identity and obtain access to the registry services.

After successful authentication, the EO/MSP verifies its identity in the EC Registry, for example through a qualified electronic signature or seal, an electronic identification means at assurance level high, or an electronic attestation. Successful verification grants the EO/MSP verified status in the registry. The EO/MSP then sets up its registry user profile and maintains the required profile data.

Once this onboarding has been completed, the EO can register battery passports directly in later registration processes. If the EO decides to use an MSP for registration, the MSP must also authenticate with the EC Registry, verify its identity, and set up its registry user profile. The EO then explicitly grants the MSP the access rights required to perform battery passport registration on its behalf.

After this setup has been completed, it is assumed that in the following user stories, the EO/MSP is verified in the EC Registry. For MSP-based registration scenarios, it is additionally assumed that the MSP is verified and that the EO has granted the MSP the required access rights.

Note: This user story defines the prerequisite process for later battery passport registration scenarios. Failure and alternative scenarios related to onboarding and access-rights setup, such as failed eIDAS authentication, rejected identity verification, incomplete profile setup, expired verified status, or missing access rights for the MSP, are not described separately here. They are considered in the related user stories for placing a battery on the market directly by the EO and for placing a battery on the market by the MSP acting on behalf of the EO.

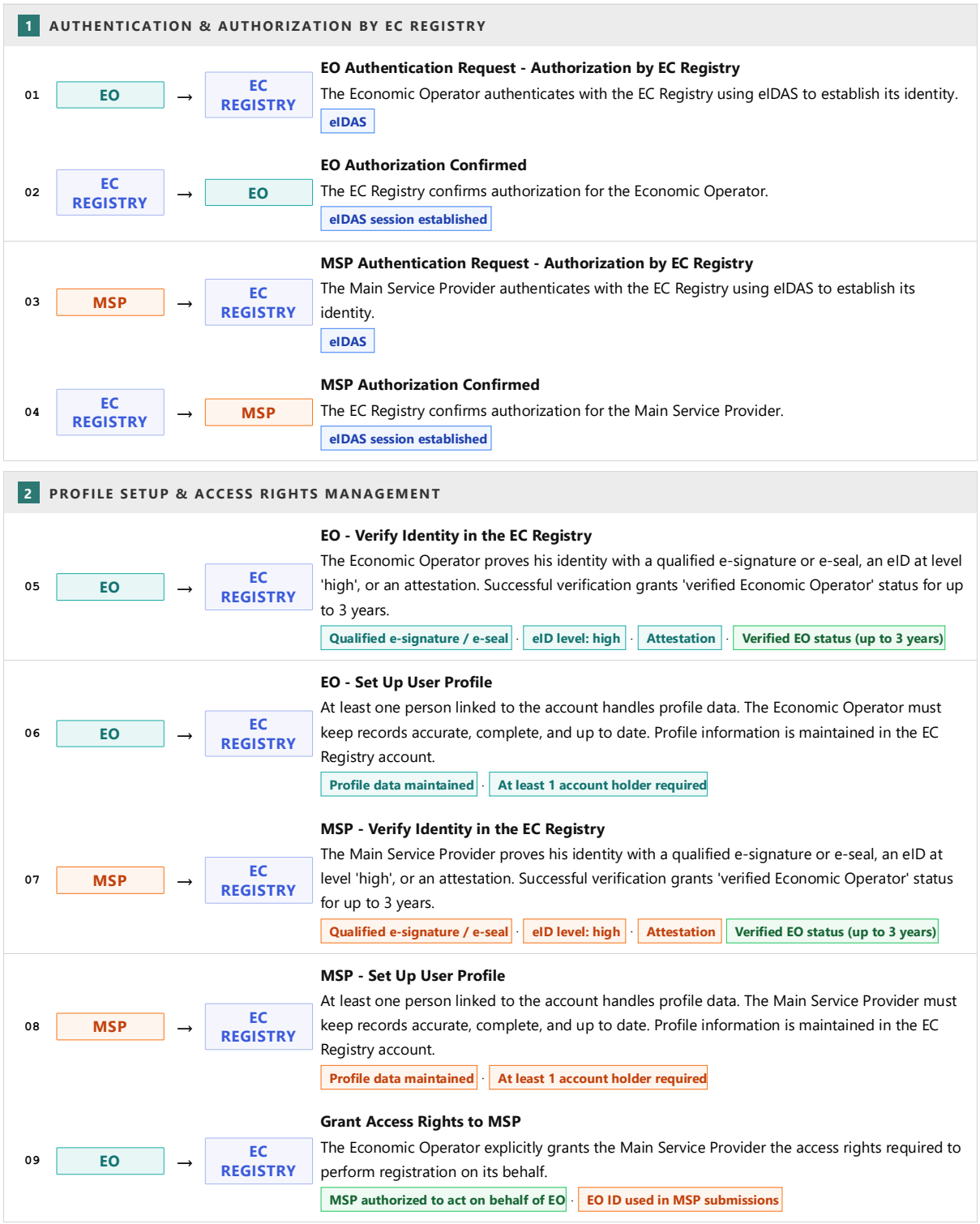


Figure 6: Ordinary Sequence "Registry Onboarding and Access Rights Granting"

6.1.2 ECONOMIC OPERATOR PLACES THE BATTERY ON THE MARKET

Overview

This user story describes how a DBP is registered in the EC Registry directly by the Economic Operator in order to place a battery on the market. It includes authentication and authorization of the Economic Operator, submission of the registration request, receipt of the Registration ID, and creation of a backup copy at the Backup Service Provider.

Assumptions

- Economic Operator authentication and authorization are performed by the EC Registry using the eIDAS framework.
- The BSP authenticates the EO using OAuth 2.0.
- The battery has not previously been registered, and all required data is available and complete at the time of registration.
- If the EC Registry receives data beyond what is required for the registration process, the registration request is rejected.
- The Registration ID is used to verify that a battery passport has been registered. It is not part of the DBP itself and is not added to the DBP data after registration.
- The CreateDPP API enables the creation of backup copies of an already existing DBP with the BSP.
- Only the DppRegistryEntry is submitted to the EC Registry during registration. The full DBP dataset is not transferred to the EC Registry as part of the registration request.

Actors and Roles

Actor	Role	Responsibilities	Authentication Protocol provided by
Economic Operator	DBP Owner/ Responsible Actor	Responsible for the end-to-end Placement on Market process. Prepares and submits the DBP data for registration and initiates backup creation with the Backup Service Provider. Receives and manages the Registration ID	
EC Registry	Registration Authority	Controls access to registry services, validates registration requests, verifies registration preconditions, registers the DBP, and issues the Registration ID.	eIDAS
Backup Service Provider	Storage Provider	Controls access to backup services, creates and stores backup copy of the DBP.	oAuth 2.0

Ordinary Sequence

The EO first authenticates and is authorized by the BSP using OAuth 2.0. After successful authentication and authorization, the EO sends the CreateDPP API call to create a backup copy of the DPP at the BSP. The BSP confirms that the backup copy has been successfully created and returns the DPP ID to the EO.

Once the backup copy has been created, the EO authenticates and is authorized by the EC Registry using eIDAS. The EO then submits the registration data either through the registry web portal or via an API call. In the API-based sequence, the EO sends a RegisterProductDPP request with the required DppRegistryEntry, including the Product ID, DPP ID, Economic Operator ID, Economic Operator ID of the Backup Service Provider, and DPP API endpoint.

The EC Registry validates the submitted data package and, if the validation is successful, generates and stores a unique Registration ID. The Registration ID is returned to the EO through the same channel used for registration and serves as confirmation that the battery passport has been successfully registered. The receipt of this Registration ID marks the successful completion of the ”Place Battery on the Market“ process.

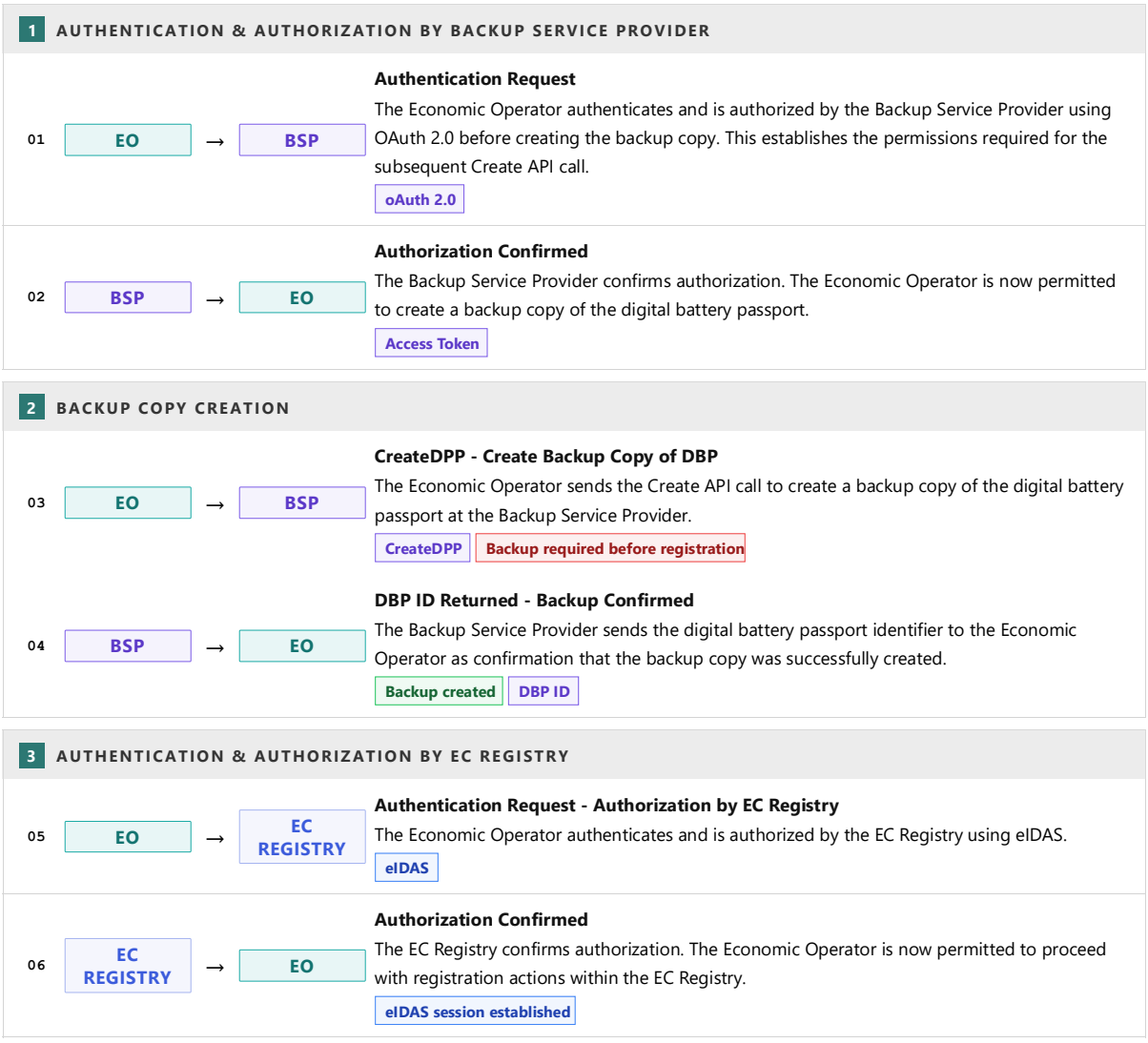


Figure 7: Ordinary Sequence "EO places the Battery on the Market. Part 1"

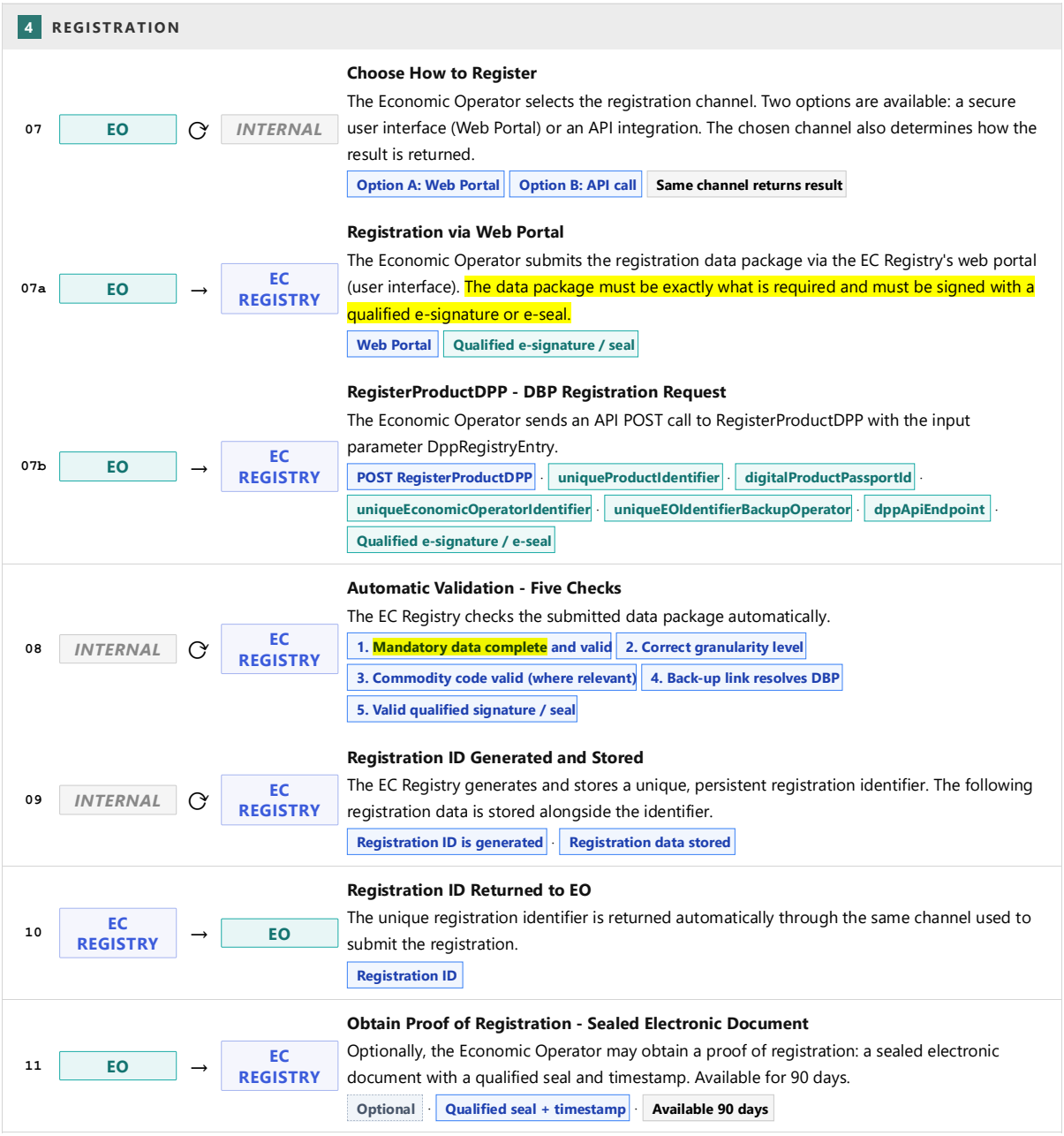


Figure 8: Ordinary Sequence "EO places the Battery on the Market. Part 2"

Failure Scenarios

The following table summarizes the failure scenarios derived from the ordinary sequence of this user story. The ID column provides a unique reference for each failure scenario. The category column groups the scenarios according to the affected aspect of the process. The failure scenario column describes the deviation from the expected process flow that the DPP system must prevent or handle correctly.

ID	Category	Failure Scenario
FS-Input-001	Input Validation	Registration request is processed although required fields in DppRegistryEntry are missing
FS-Input-002	Input Validation	Registration request is processed although fields in DppRegistryEntry are malformed or invalid

ID	Category	Failure Scenario
FS-RegAuth-001	Registry Authentication	EC Registry processes a registry action although the EO is not authenticated via eIDAS
FS-RegAuth-002	Registry Authentication	EC Registry accepts invalid, expired, revoked, malformed, or untrusted eIDAS-based identity evidence for the EO.
FS-RegAuth-003	Registry Authorization	EC Registry grants registry access to an actor who is not authorized to act as the EO
FS-RegVer-001	Registry Verification	EC Registry allows an authenticated but unverified EO to perform registry actions
FS-RegVer-002	Registry Verification	EC Registry accepts EO identity verification although the provided qualified e-signature, e-seal, eID, or attestation is missing or invalid
FS-RegVer-003	Registry Verification	EC Registry rejects EO identity verification although valid verification evidence is provided
FS-RegVer-004	Registry Verification	EC Registry does not recognize the EO as verified although the verified EO status is still within its valid three-year period
FS-RegVer-005	Registry Verification	EC Registry still treats the EO as verified although the verified EO status has expired
FS-RegProf-001	Registry Profile	EO profile setup is accepted although required profile or account holder information is missing
FS-RegProf-002	Registry Profile	EO profile data is not correctly stored, updated, or associated with the verified EO
FS-BspAuth-001	BSP Authentication	BSP processes backup creation although the EO is not authenticated via OAuth 2.0
FS-BspAuth-002	BSP Authentication	BSP accepts invalid, expired, revoked, or malformed OAuth 2.0 token
FS-BspAuth-003	BSP Authorization	Unauthorized EO is able to create a backup copy
FS-Reg-001	Registration	EC Registry rejects a valid registration request although all preconditions are fulfilled, and the battery has not been registered previously
FS-Reg-002	Registration	EC Registry accepts registration although the battery has already been registered previously
FS-Reg-003	Registration	Same registration request is processed twice, creating duplicate registrations
FS-Reg-004	Registration	Excessive data fields are accepted beyond what the DppRegistryEntry schema permits
FS-Gran-001	Granularity	EC Registry accepts registration although the referenced DBP data has an incorrect granularity level
FS-Endp-001	DPP API Endpoint	EC Registry accepts registration although the DPP API endpoint is missing, invalid, or unreachable
FS-EO-001	Economic Operator Identifier	EC Registry accepts registration although the EO ID is missing, invalid, or does not match the authenticated EO

ID	Category	Failure Scenario
FS-Bck-001	Backup	Mandatory backup copy is not created before EC Registry registration
FS-Bck-002	Backup	BSP returns success although the backup copy was not actually created
FS-Bck-003	Backup	Wrong DBP dataset is stored as backup copy
FS-Bck-004	Backup	DBP ID is not returned after successful backup creation
FS-Bck-005	Backup	Wrong DBP ID is returned as backup confirmation
FS-Bck-006	Backup	Duplicate backup copies are created for the same backup creation request
FS-Bck-007	Backup Link	EC Registry accepts registration although the backup link is missing, invalid, or cannot be resolved
FS-Bck-008	Backup / DPP Data Completeness	Backup copy is created although required battery data inside the DBP dataset is incomplete or missing
FS-ID-001	Identifier	Registration ID is not returned after successful registration
FS-ID-002	Identifier	Wrong Registration ID is returned for the registered DBP
FS-ID-003	Identifier	Non-unique Registration ID is issued
FS-Channel-001	Result Channel	Registration result is returned through the wrong channel instead of the channel used for submission
FS-Proof-001	Proof of Registration	Proof of registration is not available although less than 90 days have passed since successful registration
FS-Proof-002	Proof of Registration	Proof of registration remains available although the 90-day availability period has expired
FS-Resub-001	Resubmission	Rejected registration request creates a partial registration although no registration should be issued before all validation checks pass
FS-Sys-001	System / Audit Trail	Audit trail is missing or records the wrong actor, timestamp, request, or outcome for backup or registration steps

Test Scenarios

The following table presents the test scenarios derived from the identified failure scenarios. The test scenario column names the case to be verified, while the objective column explains the purpose of the test. The preconditions column defines the conditions that must be fulfilled before the test is executed. The expected result column describes the system behavior required for a successful test outcome, and the related failure scenario column links the test to the corresponding failure scenarios.

Test Scenario	Objective	Preconditions	Expected Result	Related Failure Scenario
EO authentication, verification, and profile setup are completed successfully	Verify that the EO can authenticate with the EC Registry, complete identity verification, and set up a valid profile	EO has valid eIDAS credentials; valid verification evidence is available; required profile and account holder information is provided	EO is authenticated, verified, and associated with a correctly stored Registry profile; verified EO status is recognized as valid	FS-RegVer-003, FS-RegVer-004, FS-RegProf-002
Mandatory backup copy is created before EC Registry registration	Verify that the DBP dataset is complete and that a correct backup copy is created before EC Registry registration	Required battery data is complete; EO is authenticated and authorized by the BSP; DBP dataset is available	Backup copy is created for the correct DBP dataset; BSP returns the correct DBP ID; exactly one intended backup copy exists	FS-Bck-001, FS-Bck-002, FS-Bck-003, FS-Bck-004, FS-Bck-005, FS-Bck-008
Valid EC Registry registration request is accepted	Verify that a valid registration request is accepted when all preconditions are fulfilled	Battery has not been registered before; backup copy exists; DppRegistryEntry is complete and valid; EO is authenticated, verified, and authorized; valid signature/seal is provided	EC Registry accepts the request, registers the DBP, returns exactly one correct and unique Registration ID through the correct channel, and the Registration ID can be used for later reference	FS-Reg-001, FS-ID-001, FS-ID-002, FS-ID-003, FS-Channel-001
Proof of registration is available within the 90-day period	Verify that the optional proof of registration can be obtained during its valid availability period	Registration was successful; less than 90 days have passed since registration	Proof of registration is available and can be retrieved within the 90-day availability period	FS-Proof-001
Audit trail is created for successful registration and backup process	Verify that all relevant process steps are traceable	EO performs authentication, backup creation, Registry registration, and receives the Registration ID	Audit trail records the correct actor, timestamp, request, outcome, DBP ID, and Registration ID where applicable	FS-Sys-001
Registration request with missing or invalid DppRegistryEntry fields is rejected	Verify that the EC Registry does not process incomplete or malformed registration input	Registration request is submitted with missing, malformed, or invalid fields in DppRegistryEntry	Request is rejected; no DBP is registered; no Registration ID is issued	FS-Input-001, FS-Input-002

Test Scenario	Objective	Preconditions	Expected Result	Related Failure Scenario
Unauthenticated or invalidly authenticated EC Registry action is rejected	Verify that EC Registry actions cannot be processed without valid eIDAS authentication	EO submits a registry action without authentication or with invalid, expired, revoked, or malformed eIDAS authentication	Request is rejected; no registry action is processed	FS-RegAuth-001, FS-RegAuth-002
Unauthorized actor cannot act as the EO in the EC Registry	Verify that only an actor authorized to act as the EO may perform registry actions	Actor is authenticated but not authorized to act as the EO	Registry access is denied; no registration or registry action is performed	FS-RegAuth-003
Authenticated but unverified EO cannot perform registry actions	Verify that eIDAS authentication alone is not sufficient and that verified EO status is required	EO is authenticated via eIDAS but is not verified, or verified EO status has expired	Registry action is rejected; no registration action is processed	FS-RegVer-001, FS-RegVer-005
Invalid EO verification evidence is rejected	Verify that EO identity verification requires valid verification evidence	EO submits missing, invalid, or unverifiable qualified e-signature, e-seal, eID, or attestation	EO verification is rejected; verified EO status is not granted	FS-RegVer-002
Invalid or incomplete EO profile setup is rejected	Verify that EO profile setup requires complete and valid profile information	EO attempts to set up a profile with missing profile or account holder information	Profile setup is rejected or remains incomplete; EO profile is not accepted as valid	FS-RegProf-001
Unauthenticated or invalidly authenticated BSP backup request is rejected	Verify that backup creation cannot be processed without valid OAuth 2.0 authentication	EO sends a backup creation request without authentication or with invalid, expired, revoked, or malformed OAuth 2.0 token	Request is rejected; no backup copy is created; no DBP ID is returned	FS-BspAuth-001, FS-BspAuth-002
Unauthorized EO cannot create a backup copy	Verify that only an authorized EO may create the backup copy	EO is not authorized to create a backup copy but calls the BSP Create API	Request is rejected; no backup copy is created	FS-BspAuth-003
Duplicate registration is	Verify that the same battery or same registration request	Battery has already been registered, or the same registration	Request is rejected or handled idempotently; no duplicate	FS-Reg-002, FS-Reg-003

Test Scenario	Objective	Preconditions	Expected Result	Related Failure Scenario
prevented or handled idempotently	cannot create duplicate registrations	request is submitted more than once	registration and no duplicate Registration ID are created	
Audit trail records correct actor and process steps	Verify traceability of authentication, registration, and backup creation	EO performs the full process or a failure occurs in one step	Audit trail records correct actor, timestamps, request, outcome, Registration ID, and DBP ID where applicable	FS-Sys-001

Note: Some of the test scenarios described in this document may not be implemented within the scope of the test environment currently being developed. This applies in particular to scenarios that depend on functions or controls expected to be implemented by the EC Registry. These scenarios are nevertheless included in this document to provide a complete overview of potential failure scenarios within the overall DPP system and to clarify their relevance for end-to-end system validation.

6.1.3 MAIN SERVICE PROVIDER PLACES THE BATTERY ON THE MARKET

Overview

This user story describes how a DBP is registered in the EC Registry when the MSP acts on behalf of the EO in order to place a battery on the market. It includes transfer of the DBP data to the MSP, registration of the battery in the EC Registry, communication of the registration result to the EO, and creation of a backup copy at the BSP.

Assumptions

- User authentication and authorization are performed by the EC Registry using the eIDAS framework.
- The MSP authenticates the EO using OAuth 2.0.
- The BSP authenticates the EO and the MSP using OAuth 2.0.
- The battery has not previously been registered, and all required data is available and complete at the time of registration.
- The Registration ID is used exclusively to verify that a battery passport has been registered. It is not part of the DPP itself and is not added to the DBP data after registration.
- If the EC Registry receives data beyond what is required for the registration process, the registration request is rejected.
- The Create API between the EO and the MSP is used to create a DBP if the MSP acts on behalf of the EO.
- The Create API between the MSP and the BSP is used as a lifecycle API to create backup copies of the DBP.
- Only the DppRegistryEntry is submitted to the EC Registry during registration. The full DBP dataset is not transferred to the EC Registry as part of the registration request.

Actors and Roles

Actor	Role	Responsibilities	Authentication Protocol provided by
Economic Operator	DBP Owner/ Legally Responsible Party	Remains legally responsible for the DBP data. Provides the DBP data to the MSP, receives confirmation of DBP creation and registration.	
Main Service Provider	Service Provider	Acts on behalf of the EO. Receives and stores the DBP data, performs the registration with the EC Registry, informs the EO of the registration result, and initiates backup creation.	OAuth 2.0
EC Registry	Registration Authority	Controls access to registry services, validates registration requests, verifies registration preconditions, registers the DBP, and issues the Registration ID.	eIDAS

Actor	Role	Responsibilities	Authentication Protocol provided by
Backup Service Provider	Storage Provider	Controls access to backup services, creates and stores the backup copy of the DBP.	OAuth 2.0

Ordinary Sequence

The EO first authenticates and is authorized by the MSP using OAuth 2.0. After successful authentication and authorization, the EO sends the CreateDPP API call with the compiled DBP data to the MSP. The MSP stores the data, creates the DBP on its server, and returns the DPP ID to the EO as initial confirmation. At this stage, the DBP has been created, but EC Registry registration is still pending.

The MSP then creates the mandatory backup copy on behalf of the EO. For this purpose, the MSP authenticates and is authorized by the BSP using OAuth 2.0. After authorization has been confirmed, the MSP sends the CreateDPP API call to the BSP to create the backup copy of the DBP. The BSP confirms that the backup copy has been successfully created and returns the DPP ID to the MSP.

Once the backup copy has been created, the MSP authenticates and is authorized by the EC Registry using eIDAS. The MSP then selects the registration channel, either the registry web portal or the API. In the API-based sequence, the MSP sends a RegisterProductDPP request with the required DppRegistryEntry, including the Product ID, DPP ID, Economic Operator ID, backup operator ID, DPP API endpoint, and qualified signature or seal. The registration package uses the EO identifiers, not the MSP identifiers.

The EC Registry validates the submitted data package and, if the validation is successful, generates and stores a unique Registration ID. The Registration ID is returned to the MSP through the same channel used for registration and serves as confirmation that the battery passport has been successfully registered. The MSP forwards the registration confirmation and the Registration ID to the EO. The receipt of this Registration ID marks the successful completion of the “Place battery on Market” process. Optionally, the MSP may obtain a sealed electronic proof of registration from the registry.

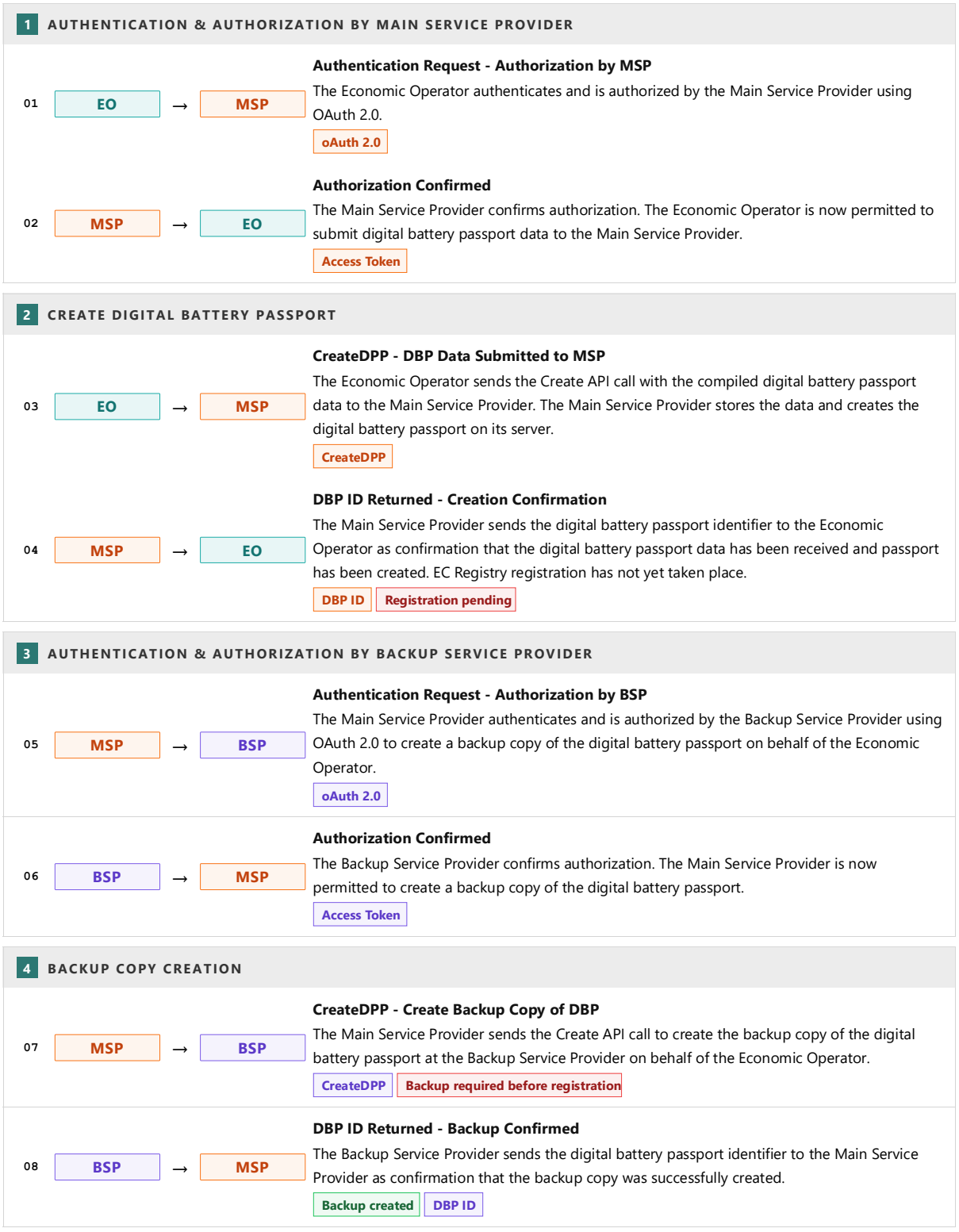


Figure 9: Ordinary Sequence "MSP places the Battery on the Market. Part 1"

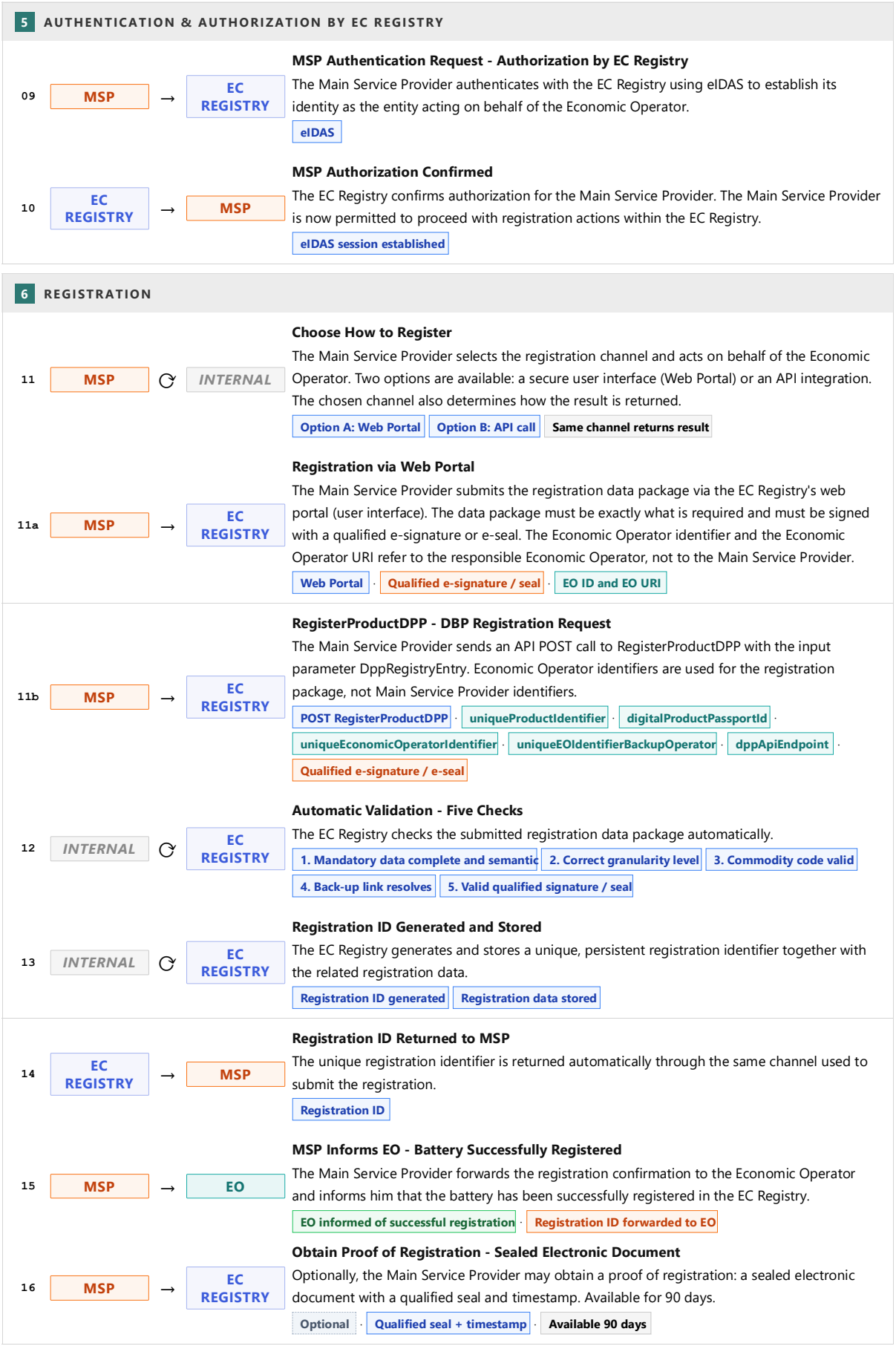


Figure 10: Ordinary Sequence "MSP places the Battery on the Market. Part 2"

Failure Scenarios

The following table summarizes the failure scenarios derived from the ordinary sequence of this user story. The ID column provides a unique reference for each failure scenario. The category column groups the scenarios according to the affected aspect of the process. The failure scenario column describes the deviation from the expected process flow that the DPP system must prevent or handle correctly.

ID	Category	Failure Scenario
FS-HndAuth-001	Handoff Authentication	MSP processes the DBP handoff request although the EO is not authenticated via OAuth 2.0
FS-HndAuth-002	Handoff Authentication	MSP accepts invalid, expired, revoked, or malformed OAuth 2.0 token from the EO
FS-HndAuth-003	Handoff Authorization	Unauthorized EO is able to hand over DBP data to the MSP
FS-Crt-001	DPP Creation on MSP Server	Valid DBP handoff request is rejected despite sufficient permissions and complete data
FS-Crt-002	DPP Creation on MSP Server	MSP stores incomplete or corrupted DBP data after handoff
FS-Crt-003	DPP Creation on MSP Server	MSP creates the DBP on its server for the wrong battery or wrong EO
FS-Crt-004	DPP Creation on MSP Server	DBP ID is not returned after successful DBP creation on the MSP server
FS-Crt-005	DPP Creation on MSP Server	Wrong DBP ID is returned as initial confirmation
FS-Crt-006	DPP Creation on MSP Server	Same DBP handoff request is processed twice, creating duplicate DBP instances on the MSP server
FS-BspAuth-001	BSP Authentication	BSP processes backup creation although the MSP is not authenticated via OAuth 2.0
FS-BspAuth-002	BSP Authentication	BSP accepts invalid, expired, revoked, or malformed OAuth 2.0 token from the MSP
FS-BspAuth-003	BSP Authorization	Unauthorized MSP is able to create a backup copy
FS-Bck-001	Backup	Mandatory backup copy is not created before EC Registry registration
FS-Bck-002	Backup	BSP returns success although the backup copy was not actually created
FS-Bck-003	Backup	Wrong DBP dataset is stored as backup copy
FS-Bck-004	Backup	DBP ID is not returned after successful backup creation
FS-Bck-005	Backup	Wrong DBP ID is returned as backup confirmation
FS-Bck-006	Backup	Same backup creation request is processed twice, creating duplicate backup copies

ID	Category	Failure Scenario
FS-Bck-007	Backup Link	EC Registry accepts registration although the backup link is missing, invalid, or cannot be resolved
FS-Bck-008	Backup / DPP Data Completeness	Backup copy is created although required battery data inside the DBP dataset is incomplete or missing
FS-RegAuth-001	Registry Authentication	EC Registry processes a registry action although the MSP is not authenticated via eIDAS
FS-RegAuth-002	Registry Authentication	EC Registry accepts invalid, expired, revoked, or malformed eIDAS authentication for the MSP
FS-RegAuth-003	Registry Authorization	EC Registry grants registry access to an actor who is not authorized to act as the EO
FS-RegVer-001	Registry Verification	EC Registry allows an authenticated but unverified MSP to perform registry actions
FS-RegVer-002	Registry Verification	EC Registry accepts MSP identity verification although the provided qualified e-signature, e-seal, eID, or attestation is missing or invalid
FS-RegVer-003	Registry Verification	EC Registry rejects MSP identity verification although valid verification evidence is provided
FS-RegVer-004	Registry Verification	EC Registry does not recognize the MSP as verified although the verified status is still within its valid three-year period
FS-RegVer-005	Registry Verification	EC Registry still treats the MSP as verified although the verified status has expired
FS-RegProf-001	Registry Profile	MSP profile setup is accepted although required profile or account holder information is missing
FS-RegProf-002	Registry Profile	MSP profile data is not correctly stored, updated, or associated with the verified MSP
FS-Del-001	Delegation / Authorization	EC Registry processes registration although the EO has not granted the MSP the right to act on its behalf
FS-Del-002	Delegation / Authorization	EC Registry processes registration although the MSP's delegated rights are expired, revoked, or invalid
FS-Del-003	Delegation / Identifier Use	EC Registry accepts registration although the MSP uses its own identifier instead of the EO identifier
FS-Del-004	Delegation / Identifier Use	EC Registry accepts registration although the EO ID or EO-related endpoint does not belong to the EO on whose behalf the MSP acts
FS-Input-001	Input Validation	Registration request is processed although required fields in DppRegistryEntry are missing
FS-Input-002	Input Validation	Registration request is processed although fields in DppRegistryEntry are malformed or invalid
FS-Reg-001	Registration	EC Registry rejects a valid registration request submitted by the MSP although all preconditions are fulfilled and the battery has not been registered previously

ID	Category	Failure Scenario
FS-Reg-002	Registration	EC Registry accepts registration although the battery has already been registered previously
FS-Reg-003	Registration	Same registration request is processed twice, creating duplicate registrations
FS-Reg-004	Registration	Excessive data fields are accepted beyond what the DppRegistryEntry schema permits
FS-Reg-005	Registration	EC Registry returns success although the DBP was not actually registered
FS-Sig-001	Signature / Seal	Registration request submitted by the MSP is accepted although the required qualified e-signature or e-seal is missing, invalid, or cannot be verified
FS-Gran-001	Granularity	EC Registry accepts registration although the referenced DBP data has an incorrect granularity level
FS-Endp-001	DPP API Endpoint	EC Registry accepts registration although the DPP API endpoint is missing, invalid, or unreachable
FS-ID-001	Identifier	Registration ID is not returned after successful registration
FS-ID-002	Identifier	Wrong Registration ID is returned for the registered DBP
FS-ID-003	Identifier	Duplicate or non-unique Registration ID is issued
FS-Channel-001	Result Channel	Registration result is returned through the wrong channel instead of the channel used for submission
FS-Proof-001	Proof of Registration	Proof of registration is not available although less than 90 days have passed since successful registration
FS-Proof-002	Proof of Registration	Proof of registration remains available although the 90-day availability period has expired
FS-Notify-001	Notification to EO	MSP does not inform the EO after successful registration
FS-Notify-002	Notification to EO	MSP forwards an incorrect Registration ID or incorrect registration confirmation to the EO
FS-Resub-001	Resubmission	Rejected registration request creates a partial registration although no registration should be issued before all validation checks pass
FS-Sys-001	System / Audit Trail	Audit trail is missing or records the wrong actor, timestamp, request, outcome, DPP ID, or Registration ID for handoff, backup, registration, notification, or identifier issuance steps

Test Scenarios

The following table presents the test scenarios derived from the identified failure scenarios. The test scenario column names the case to be verified, while the objective column explains the purpose of the test. The preconditions column defines the conditions that must be fulfilled before the test is executed. The expected result column describes the system behavior required for a successful test outcome, and

the related failure scenario column links the test to the corresponding failure scenarios.

Test Scenario	Objective	Preconditions	Expected Result	Related Failure Scenario
Valid DBP handoff to MSP is accepted and processed correctly	Verify that a valid DBP handoff request is accepted, stored correctly, and confirmed with the correct DBP ID	EO is authenticated and authorized by the MSP; DBP data is complete and valid; battery and EO data are correct	MSP accepts the handoff, stores the complete and unchanged DBP dataset, creates the DBP for the correct battery and EO, and returns exactly one correct DBP ID	FS-Crt-001, FS-Crt-004, FS-Crt-005
Mandatory backup copy is created before EC Registry registration	Verify that the MSP creates a valid backup copy before submitting the EC Registry registration request	DBP has been created on the MSP server; MSP is authenticated and authorized by the BSP; DBP dataset is complete and valid	Backup copy is created for the correct DBP dataset before EC Registry registration; BSP returns the correct DBP ID as backup confirmation	FS-Bck-001, FS-Bck-002, FS-Bck-003, FS-Bck-004, FS-Bck-005, FS-Bck-008
MSP authentication, verification, and profile setup completed successfully	Verify that the MSP can be authenticated, verified, and correctly associated with its EC Registry profile	MSP provides valid eIDAS authentication and valid verification evidence; required profile and account holder information is complete	EC Registry authenticates the MSP, recognizes it as verified, stores the MSP profile correctly, and associates it with the verified MSP	FS-RegVer-003, FS-RegVer-004, FS-RegProf-002
Valid delegation from EO to MSP is recognized by the EC Registry	Verify that the MSP is allowed to act on behalf of the EO only when valid delegation exists	EO is authenticated and authorized to grant rights; MSP is verified; delegation rights are valid and active	EC Registry recognizes the MSP as authorized to act on behalf of the EO	FS-Del-001, FS-Del-002
Valid EC Registry registration request submitted by MSP is accepted	Verify successful registration when all preconditions are fulfilled	Battery has not been registered before; DppRegistryEntry is complete and valid; backup link is valid; MSP is authenticated, verified, and authorized to act on behalf of the EO	EC Registry accepts the request, registers the DBP, returns exactly one correct Registration ID, and sends the result through the correct channel	FS-Reg-001, FS-Reg-005, FS-ID-001, FS-ID-002, FS-ID-003, FS-Channel-001
Proof of registration is	Verify that proof of registration can be	Registration has been completed	Proof of registration is available and	FS-Proof-001

Test Scenario	Objective	Preconditions	Expected Result	Related Failure Scenario
available within the 90-day period	accessed within the allowed availability period	successfully; less than 90 days have passed since successful registration	refers to the correct registered DBP	
EO is informed after successful registration by the MSP	Verify that the EO receives the correct registration confirmation from the MSP	Registration has been completed successfully by the MSP on behalf of the EO	MSP informs the EO and forwards the correct Registration ID and registration confirmation	FS-Notify-001, FS-Notify-002
Audit trail is created for the complete MSP registration process	Verify traceability of handoff, backup, registration, notification, and identifier issuance	Full registration process is executed successfully, or a controlled failure occurs in one process step	Audit trail records the correct actor, timestamp, request, outcome, DBP ID, Registration ID, and relevant process step	FS-Sys-001
Unauthenticated, invalidly authenticated, or unauthorized DBP handoff to MSP is rejected	Verify that the MSP only accepts DBP handoff from a properly authenticated and authorized EO	EO or another actor sends the DBP handoff request without authentication, with an invalid/expired/revoked/malformed OAuth 2.0 token, or without required authorization	Request is rejected; no DBP data is stored; no DBP is created on the MSP server	FS-HndAuth-001, FS-HndAuth-002, FS-HndAuth-003
DBP handoff with incomplete, corrupted, or wrong data is not stored as valid DBP	Verify that the MSP does not create or store an incorrect DBP after handoff	Handoff request contains incomplete, corrupted, or inconsistent DBP data, or data belonging to the wrong battery or EO	Request is rejected; no incorrect DBP instance is created on the MSP server	FS-Crt-002, FS-Crt-003
Duplicate DBP handoff request is rejected	Verify that the same handoff request cannot create duplicate DBP instances on the MSP server	A valid DBP handoff request has already been processed; the same request is submitted again	No unintended second DBP instance is created; repeated request is rejected or handled safely according to idempotency rules	FS-Crt-006
Unauthenticated or invalidly authenticated BSP request from MSP is rejected	Verify that backup creation cannot be processed without valid OAuth 2.0 authentication	MSP calls BSP Create API without valid authentication or with an	Request is rejected; no backup copy is created; no DBP ID is returned	FS-BspAuth-001, FS-BspAuth-002

Test Scenario	Objective	Preconditions	Expected Result	Related Failure Scenario
		invalid/expired/revoked/malformed OAuth 2.0 token		
Unauthorized MSP cannot create backup copy	Verify that only the authorized MSP may create the backup copy	Actor is not authorized for backup creation but calls BSP Create API	Request is rejected; no backup copy is created	FS-BspAuth-003
Duplicate backup creation request is rejected or handled idempotently	Verify that repeating the same backup creation request does not create duplicate backup copies	A valid backup creation request has already been processed; the same request is submitted again	Only one backup copy exists, or repeated calls are handled safely according to idempotency rules	FS-Bck-006
Backup creation with incomplete DBP dataset is rejected	Verify that the BSP does not create a backup copy for incomplete battery or product data	MSP submits a backup creation request with missing or incomplete required battery data inside the DBP dataset	Backup creation is rejected; no valid backup confirmation is returned	FS-Bck-008
Registration with missing, invalid, or unresolved backup link is rejected	Verify that EC Registry registration requires a valid and resolvable backup link	MSP submits a registration request with a missing, invalid, or non-resolvable backup link	Registration request is rejected; no Registration ID is issued	FS-Bck-007
Unauthenticated or invalidly authenticated EC Registry request from MSP is rejected	Verify that the EC Registry does not process registry actions without valid eIDAS authentication	MSP submits a registry request without authentication or with invalid/expired/revoked/malformed eIDAS credentials	Request is rejected; DBP is not registered; no Registration ID is issued	FS-RegAuth-001, FS-RegAuth-002
Actor not authorized to act as MSP cannot access EC Registry functions	Verify that EC Registry access is only granted to actors authorized to act as the EO	Actor submits a registry request but is not authorized to act as the EO	EC Registry access is denied; no registry action is processed	FS-RegAuth-003
Authenticated but unverified or expired MSP	Verify that successful authentication alone is not	MSP is authenticated but not verified, or the verified MSP	EC Registry rejects the registry action; no registration is processed	FS-RegVer-001, FS-RegVer-005

Test Scenario	Objective	Preconditions	Expected Result	Related Failure Scenario
cannot perform registry actions	sufficient for registry actions	status has expired		
Invalid MSP verification evidence or incomplete MSP profile setup is rejected	Verify that MSP verification and profile setup require valid evidence and complete profile information	MSP provides missing or invalid verification evidence, or required profile/account holder information is missing	Verification or profile setup is rejected; MSP is not recognized as valid verified MSP for registry actions	FS-RegVer-002, FS-RegProf-001
Missing, expired, revoked, or invalid EO delegation is rejected	Verify that MSP registration on behalf of the EO requires valid delegation rights	MSP attempts to register a DBP on behalf of the EO without valid delegation, or with expired/revoked/invalid delegated rights	Registration request is rejected; no registration takes place	FS-Del-001, FS-Del-002
Registration with wrong EO ID or EO-related endpoint is rejected	Verify that the MSP uses the correct EO ID and EO-related endpoint when acting on behalf of the EO	MSP submits a registration request using its own identifier instead of the EO ID, or using an EO-related endpoint that does not belong to the EO	Registration request is rejected; no Registration ID is issued	FS-Del-003, FS-Del-004
Registration request with missing or malformed DppRegistryEntry fields is rejected	Verify that the EC Registry only processes complete and valid DppRegistryEntry data	MSP submits a registration request with missing required fields or malformed/invalid fields in DppRegistryEntry	Registration request is rejected; no registration is created; no Registration ID is returned	FS-Input-001, FS-Input-002
Registration request with schema, signature, granularity, or endpoint errors is rejected	Verify that EC Registry registration rejects invalid registration packages	DppRegistryEntry contains excessive fields, invalid signature/seal information, incorrect granularity, or an invalid/unreachable DPP API endpoint	Registration request is rejected; no Registration ID is issued	FS-Reg-004, FS-Sig-001, FS-Gran-001, FS-Endp-001

Test Scenario	Objective	Preconditions	Expected Result	Related Failure Scenario
Duplicate registration is rejected or handled idempotently	Verify that the same battery or same registration request cannot create duplicate registrations	Battery has already been registered previously, or the same registration request is submitted again	Request is rejected or handled safely according to idempotency rules; no second registration is created	FS-Reg-002, FS-Reg-003
EC Registry returns success only if the DBP is actually registered	Verify that the EC Registry does not report successful registration if the DBP was not actually registered	MSP submits a registration request and the system response is checked against the actual EC Registry state	Success is returned only if the DBP is truly registered and linked to the correct battery	FS-Reg-005
Proof of registration is no longer available after the 90-day period	Verify that proof of registration is only available within the defined availability period	Registration has been completed successfully; more than 90 days have passed since successful registration	Proof of registration is no longer available	FS-Proof-002
Rejected registration request does not create partial registration	Verify that failed registration attempts do not create partial or inconsistent registry entries	MSP submits an invalid registration request that is rejected by the EC Registry	No partial registration is created; no Registration ID is issued before all validation checks pass	FS-Resub-001

Note: Some of the test scenarios described in this document may not be implemented within the scope of the test environment currently being developed. This applies in particular to scenarios that depend on functions or controls expected to be implemented by the EC Registry service provider. These scenarios are nevertheless included in this document to provide a complete overview of potential failure scenarios within the overall DPP system and to clarify their relevance for end-to-end system validation.

6.2 ACCESS BATTERY PASSPORT DATA

6.2.1 READING PUBLIC BATTERY PASSPORT DATA

Overview

This user story describes how all actors within the DBP ecosystem can access specific battery passport data of public interest using a QR code affixed to the battery or included within the battery documentation. Access to publicly available data shall be enabled via a mobile device capable of scanning the QR code (e.g., a smartphone). Upon scanning the QR code, the user shall be directed to a web-based interface where the publicly accessible battery passport data is rendered in a structured, human-readable format on an HTML page. The solution shall ensure seamless redirection from the QR code to the corresponding online resource and shall provide public accessibility without authentication barriers for data classified as public. The DBP information shall be presented in a clear and user-friendly manner and shall be compatible with standard mobile browsers and commonly used scanning applications.

Actors and Roles

Actor	Role	Responsibilities	Authentication Protocol provided by
Economic Operator/ Main Service Provider	DBP Data Provider / Website Host	Publishes and maintains publicly accessible battery passport data. Ensures that the QR code resolves to a persistent URL. Hosts the web interface rendering public DBP data in human-readable HTML format. Ensures data accuracy, availability, and separation between public and restricted data domains.	No authentication required for public data access
End User (e.g., Consumer)	Public Data Viewer	Scans QR code using a mobile device. Accesses the redirected web interface. Views publicly available battery passport data.	

Ordinary Sequence

The EO labels the product with a QR code that contains a Product ID/ Battery ID and a direct URL to the publicly accessible DBP data hosted online.

When an end user scans the QR code with a mobile device, the encoded URL is extracted and a secure request is automatically sent to the EO/MSP Server. The server processes the request, identifies the requested information as public data, and retrieves the corresponding battery passport information from its data repository without requiring authentication.

The data is then displayed as a structured, human-readable HTML page optimized for mobile browsers, allowing the end user to review the publicly available battery passport information without any access barriers.

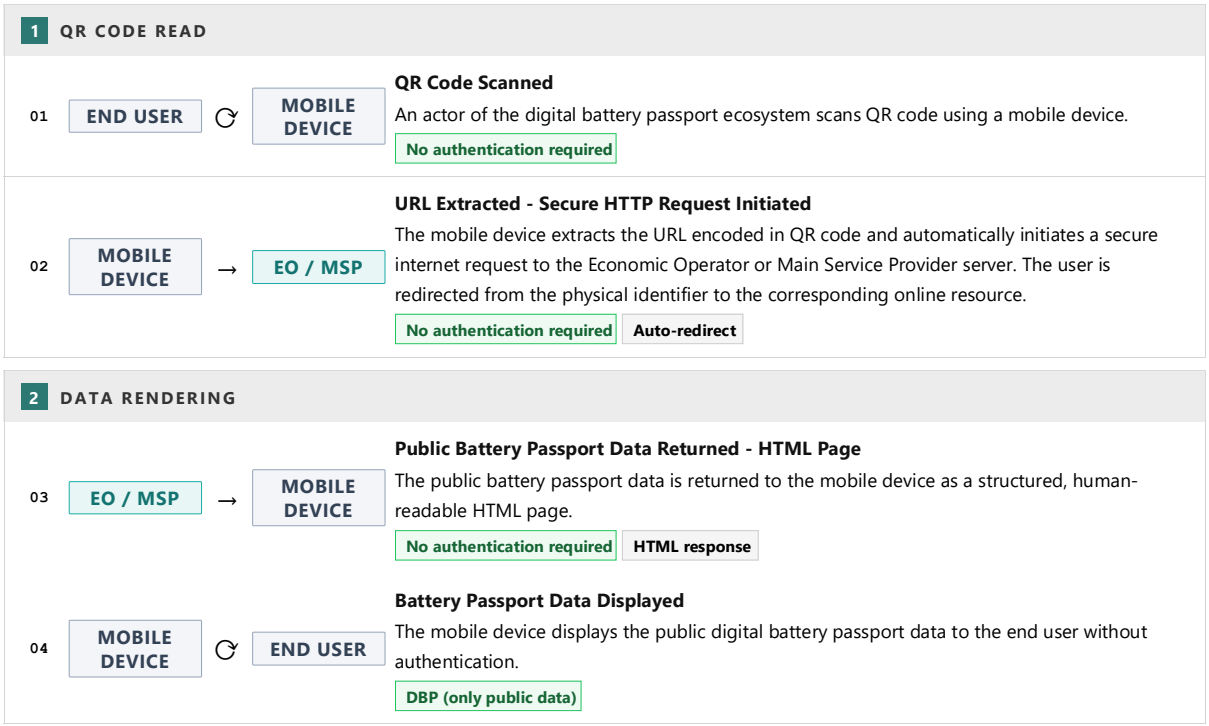


Figure 11: Ordinary Sequence "Reading Public Battery Passport Data"

Failure Scenarios

The following table summarizes the failure scenarios derived from the ordinary sequence of this user story. The ID column provides a unique reference for each failure scenario. The category column groups the scenarios according to the affected aspect of the process. The failure scenario column describes the deviation from the expected process flow that the DPP system must prevent or handle correctly.

ID	Category	Description
FS-QR-001	QR Code	QR-Code is missing on the product or product documentation
FS-QR-002	QR Code	QR-Code is damaged and cannot be scanned correctly
FS-QR-003	QR Code	QR-Code contains an invalid, malformed, or incomplete URL
FS-QR-004	QR Code	QR-Code is not connected to a functional link
FS-QR-005	QR Code	QR-Code is connected to wrong battery passport
FS-QR-006	QR Code	QR-Code points to a non-persistent or expired URL
FS-Reg-001	Registration / Availability	Battery passport is not registered
FS-Red-001	Redirection	Scanning the QR-Code does not trigger redirection to the public battery passport page
FS-Red-002	Redirection	Redirection leads to an HTTP error page although the battery passport is registered and public data is available

ID	Category	Description
FS-Acc-001	Access	Public battery passport data is blocked by an authentication or login barrier
FS-Acc-002	Access	Public page is unavailable due to host outage or intermittent service failure
FS-Data-001	Data Retrieval	Public battery passport data exists but is not returned by the host system
FS-Data-002	Data Retrieval	Public page displays outdated battery passport data instead of the current public version
FS-Data-003	Data Retrieval	Public page displays incomplete public data
FS-Data-004	Data Retrieval	Public and restricted data domains are not correctly separated
FS-Priv-001	Privacy / Access Control	Restricted or controlled data is exposed on the public HTML page
FS-Rend-001	Rendering	Public battery passport data is not rendered in a structured, human-readable HTML format
FS-Rend-002	Rendering	HTML page is unreadable or poorly formatted on a standard mobile device
FS-Sys-001	System	Cached response exposes outdated or incorrect public data after updates
FS-Sys-002	System	Different devices or browsers display inconsistent public battery passport content

Test Scenarios

The following table presents the test scenarios derived from the identified failure scenarios. The test scenario column names the case to be verified, while the objective column explains the purpose of the test. The preconditions column defines the conditions that must be fulfilled before the test is executed. The expected result column describes the system behavior required for a successful test outcome, and the related failure scenario column links the test to the corresponding failure scenarios.

Test Scenario	Objective	Preconditions	Expected Result	Related Failure Scenario
QR code enables valid and correct access to the public battery passport	Verify that the QR code is physically available, scannable, technically valid, functional, persistent, and connected to the correct battery passport	Battery or battery documentation contains a QR code; battery has an assigned public battery passport	QR code is present, visible, undamaged, and can be scanned successfully. The encoded URL is valid, complete, functional, persistent, and opens the correct battery passport page for the corresponding battery only	FS-QR-001, FS-QR-002, FS-QR-003, FS-QR-004, FS-QR-005, FS-QR-006

Test Scenario	Objective	Preconditions	Expected Result	Related Failure Scenario
Unregistered battery passport cannot be accessed as valid public passport	Verify system behavior when a QR code points to a battery passport that is not registered	Battery ID exists, but no battery passport is registered for it	System does not display a valid public battery passport. Instead, an appropriate error, not-found, or invalid passport page is shown	FS-Reg-001
QR scan redirects to an available public battery passport page without authentication	Verify that scanning the QR code redirects the user to an available public battery passport page without login or access barriers	Valid QR code exists; battery passport is registered; public page is hosted and should be available	Scanning the QR code triggers redirection to the public battery passport page. The page opens without authentication, login, token, registration, access approval, HTTP error, timeout, or host failure	FS-Red-001, FS-Red-002, FS-Acc-001, FS-Acc-002
Public battery passport page returns current and complete public data	Verify that the host system returns the available public battery passport data completely and in its current version	Public battery passport data exists in the host system; a current public version is available	Host system returns the public battery passport data. The displayed data is complete, corresponds to the current public version, and is not outdated, incorrect, empty, or affected by cached responses	FS-Data-001, FS-Data-002, FS-Data-003, FS-Sys-001
Public page exposes only public data and no controlled data	Verify that public and controlled data domains are correctly separated on the public HTML page	Battery passport contains both public and controlled data fields	Public page displays only public data. Controlled data is not visible on the page, not included in the HTML source, and not returned through the public request	FS-Data-004, FS-Priv-001
Public battery passport page is rendered consistently and readably across supported devices and browsers	Verify that the public battery passport is displayed as structured, human-readable HTML and remains readable and consistent across supported environments	Public battery passport data is available and accessible; page is opened on supported mobile devices and browsers	Data is rendered as structured, human-readable HTML. The page is readable and properly formatted on a standard mobile device, and all supported devices and browsers display the same public battery passport content consistently	FS-Rend-001, FS-Rend-002, FS-Sys-002

Note: Some of the test scenarios described in this document may not be implemented within the scope of the test environment currently being developed. These scenarios are nevertheless included in this document to provide a complete overview of potential failure scenarios within the overall DPP system and to clarify their relevance for end-to-end system validation.

6.2.2 READING CONTROLLED BATTERY PASSPORT DATA

Overview

This user story describes how an authenticated actor retrieves controlled data from a digital battery passport based according to their assigned access rights. Controlled data is not publicly accessible and is therefore protected by an authentication and authorization process managed by the Economic Operator or Main Service Provider. Once the actor's identity and permissions have been verified, the system returns the data that the actor is permitted to see.

Assumptions

- The EO/MSP authenticates downstream actors working with the EO using OAuth 2.0.
- The EO/MSP authenticates market surveillance authorities, customs authorities, other competent authorities, and regulated base roles using the eIDAS framework.
- An actor's read permissions always include all data for which the actor has read-write permissions and additionally include data that is available in read-only mode.
- Public battery passport data must always be visible to all actors, regardless of their individual access rights.
- When a DBP is retrieved using readDPPByProductId (Battery ID), the active version of the requested DBP is returned.
- When a DBP is retrieved using readDPPById, the corresponding version of the requested DBP is returned regardless of its status: active, archived, or marked for deletion.

Actors and Roles

Actor	Role	Responsibilities	Authentication Protocol provided by
Economic Operator / Main Service Provider	DBP Data Provider	Hosts the DBP and manages access rights for all actors. Authenticates and authorizes incoming read requests. Returns only the data the requesting actor is permitted to see, filtered by their access rights. Enforces the distinction between public and controlled data fields.	OAuth2.0 or eIDAS
Third party actor Market surveillance authorities and other public bodies	Requesting actor	Initiates the read. Submits a GET request using one of the four defined read functions. Receives only the data fields their access rights permit.	

Ordinary Sequence

Actors authenticate and are authorized by the Economic Operator. Depending on the actor's role, authentication is performed using OAuth 2.0 for third party actors, or eIDAS in cases involving market surveillance authorities and other public bodies requiring a higher assurance level. After successful authentication and authorization, the actor receives an access token that encodes their role-based access rights. Using the access token, the actor submits a GET request to the EO/MSP, selecting the applicable read function and providing the required input parameters. The choice of function depends on what the actor wants to retrieve: a full DBP by its ID, a DBP by Battery ID, a historical version by timestamp and DBP ID, or an individual data element by its path. Upon receiving the request, the EO checks whether the actor has the necessary access rights. If access is granted, the system displays the relevant battery passport data in accordance with those rights.

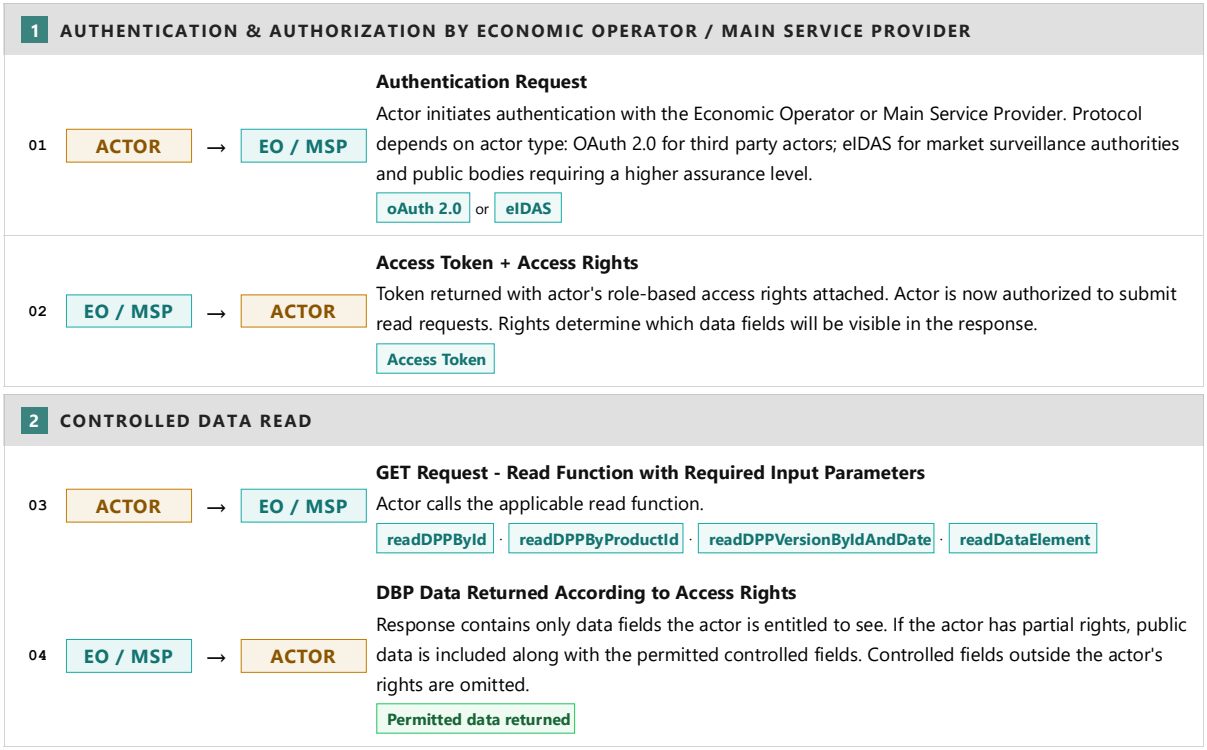


Figure 12: Ordinary Sequence "Reading Controlled Battery Passport Data"

Failure Scenarios

The following table summarizes the failure scenarios derived from the ordinary sequence of this user story. The ID column provides a unique reference for each failure scenario. The category column groups the scenarios according to the affected aspect of the process. The failure scenario column describes the deviation from the expected process flow that the DPP system must prevent or handle correctly.

ID	Category	Failure Scenario
FS-Input-001	Input Validation	Read request is processed although required input parameters are missing

ID	Category	Failure Scenario
FS-Auth-001	Authentication	Read request is processed although the actor is not authenticated
FS-Auth-002	Authentication	DPP system accepts an invalid, expired, or revoked access token
FS-Auth-003	Authentication	DPP system grants an unauthorized actor access to controlled battery passport data
FS-Auth-004	Authentication / Access Control	Actor without read rights accesses a controlled data element
FS-Auth-005	Authorization / Tenant Boundary	Actor accesses battery passport data of another Economic Operator without permission, while calling an MSP that serves multiple Economic Operators
FS-Read-001	Read	Valid read request is rejected despite sufficient permissions
FS-Read-002	Read	readDPPById returns the wrong DBP for the given DBP ID
FS-Read-003	Read	readDPPByProductId returns a DBP for the wrong Product ID/ Battery ID
FS-Read-004	Read / Versioning	readDPPVersionByIdAndDate returns the wrong DBP version for the given date
FS-Read-005	Read	APIs return an error although the requested DBP data is available
FS-Read-006	Read / Access Control	Response contains controlled fields that the actor is not permitted to see
FS-Read-007	Read / Data Completeness	The read request is successful, but the returned battery passport data is incomplete for the requesting actor’s access rights
FS-Read-008	Read	API returns DBP data although the requested DBP does not exist
FS-Acc-001	Access Control	System displays data beyond the actor’s access rights
FS-Acc-002	Access Control	System hides public data although the actor is allowed to view it
FS-Acc-003	Access Control	readDataElement returns a controlled element despite missing access rights
FS-Ver-001	Versioning	readDPPByProductId returns an outdated version instead of the current active version
FS-Ver-002	Versioning	readDPPVersionByIdAndDate returns no version although a valid version exists for the given date
FS-Ver-003	Versioning	readDPPVersionByIdAndDate returns the latest version instead of the version valid at the requested timestamp
FS-Data-001	Data Retrieval	readDataElement resolves the wrong element ID path
FS-Data-002	Data Retrieval	Returned data element belongs to a different DBP than requested

ID	Category	Failure Scenario
FS-Sys-001	System / Audit Trail	Audit trail is missing or records the wrong actor for read access

Test Scenarios

The following table presents the test scenarios derived from the identified failure scenarios. The test scenario column names the case to be verified, while the objective column explains the purpose of the test. The preconditions column defines the conditions that must be fulfilled before the test is executed. The expected result column describes the system behavior required for a successful test outcome, and the related failure scenario column links the test to the corresponding failure scenarios.

Test Scenario	Objective	Preconditions	Expected Result	Related Failure Scenario
readDPPById returns the requested DBP	Verify that readDPPById successfully returns the correct DBP for a valid DBP ID	DBP exists; actor is authenticated and authorized; valid DBP IDis provided	Request succeeds; response contains exactly the DBP belonging to the requested DBP ID; no false API error is returned	FS-Read-001, FS-Read-002, FS-Read-005
readDPPByProductId returns the current active DBP	Verify that readDPPByProductId successfully returns the correct current active DBP for a valid Battery ID	Battery exists; battery has an active DBP version; actor is authenticated and authorized; valid Battery ID is provided	Request succeeds; response contains the DBP belonging to the requested Battery ID and returns the current active version; no outdated version is returned	FS-Read-001, FS-Read-003, FS-Read-005, FS-Ver-001
readDPPVersionByIdAndDate returns the version valid at the requested date	Verify that readDPPVersionByIdAndDate successfully returns the DBP version valid at the requested timestamp	DBP has multiple historical versions; actor is authenticated and authorized; valid DBP ID and valid date are provided	Request succeeds; response contains the DBP version valid at the requested timestamp, not the latest version and not an empty result	FS-Read-001, FS-Read-004, FS-Read-005, FS-Ver-002, FS-Ver-003
readDataElement returns the requested permitted data element	Verify that readDataElement successfully returns the correct data element from the requested DBP according to the actor's access rights	DBP exists; requested data element exists; actor is authenticated and authorized for the requested element	Request succeeds; response contains only the requested authorized data element from the correct DBP; no restricted or unrelated element is returned	FS-Read-001, FS-Read-005, FS-Data-001, FS-Data-002, FS-Acc-003
Read request with missing required	Verify that the system does not process read	DBP exists; actor is authenticated and	Request is rejected with an appropriate client error response;	FS-Input-001

Test Scenario	Objective	Preconditions	Expected Result	Related Failure Scenario
input parameters is rejected	requests if mandatory input parameters are missing	authorized; one or more required input parameters are missing	no DBP data is returned	
Unauthenticated or invalid-token read request is rejected	Verify that controlled read access is blocked if the actor is not authenticated or presents an invalid, expired, or revoked token	DBP exists; actor sends a read request either without authentication or with an invalid, expired, or revoked token	Request is rejected with 401 ClientNotAuthorized or 403 ClientForbidden; no controlled data is returned	FS-Auth-001, FS-Auth-002
Authenticated but unauthorized actor cannot access controlled DBP data	Verify that authentication alone is not sufficient and that only actors with valid read rights may access controlled data	Actor is successfully authenticated, but has no authorization for controlled DBP data	Request is rejected or returns no controlled content	FS-Auth-003, FS-Acc-001
Actor without element-level read rights cannot access controlled data element	Verify that element-level access control is enforced for controlled data elements	Actor is authenticated, but lacks read rights for the requested controlled data element	Request is rejected or the controlled data element is not returned	FS-Auth-004, FS-Acc-001, FS-Acc-003
Actor cannot access DBP data belonging to another EO or tenant	Verify that tenant and ownership boundaries are enforced during controlled read access	Actor is authenticated and may have read rights in one tenant or EO context, but requests DBP data belonging to another EO, or tenant	Request is rejected or no data from the other EO or tenant is returned	FS-Auth-005
Request for nonexistent DBP or battery does not return DBP data	Verify that the APIs readDPPbyProductID and readDPPbyID do not return DBP data if the requested DBP or battery does not exist	Actor is authenticated and authorized; requested DBP ID or Battery ID does not exist in the system	Request returns an appropriate not-found or invalid-reference response; no DBP data is returned	FS-Read-008
Response is filtered exactly to	Verify that the system neither exposes data beyond the	DBP contains public fields and controlled fields; actor has	Response contains all fields the actor is allowed to see, including public data, and	FS-Read-006, FS-Read-007,

Test Scenario	Objective	Preconditions	Expected Result	Related Failure Scenario
the actor's access rights	actor's permissions nor omits data the actor is allowed to see	limited but valid read rights	excludes all fields outside the actor's access scope	FS-Acc-001, FS-Acc-002
Controlled read access is logged with the correct actor	Verify that traceability for controlled read access is ensured	Actor performs a controlled read on an existing DBP	Audit trail is created and records the correct actor, requested object, timestamp, and outcome	FS-Sys-001

Note: Some of the test scenarios described in this document may not be implemented within the scope of the test environment currently being developed. These scenarios are nevertheless included in this document to provide a complete overview of potential failure scenarios within the overall DPP system and to clarify their relevance for end-to-end system validation.

6.3 SEARCH FOR BATTERY PASSPORTS

Overview

This user story describes the search process within the DBP ecosystem using the ReadDPPIdsByProductIds method. Search functionality is needed when an authorized actor does not yet know the relevant DBP ID but has one or more Battery IDs and needs to discover which DBPs are linked to them.

Note: The JTC 24 API specifications mention several methods that may support search-related functionality. However, not all of them are treated as separate search processes in this document. ReadDPPByProductId is understood as a regular read function because it directly returns the corresponding DBP for a known Battery ID. It is therefore covered in the user story on reading controlled battery passport data. A possible read function based on Product ID (Battery ID) and date is not considered here because the corresponding API endpoint and input parameters are not defined in the JTC 24 standards.

Assumptions

- The EO/MSP authenticates downstream actors working with the EO using OAuth 2.0.
- The EO/MSP authenticates market surveillance authorities, customs authorities, other competent authorities, and regulated base roles using the eIDAS framework.
- The ReadDPPIdsByProductIds function does not provide access to DBP content, but it reveals the linkage between Battery IDs and DBP IDs. It is therefore treated as a controlled discovery function and is available only to authenticated and authorized actors with the corresponding search permission.

Actors and Roles

Actor	Role	Responsibilities	Authentication Protocol provided by
Economic Operator / Main Service Provider	DBP Data Provider	Hosts the DBP and manages access rights. Authenticates and authorizes incoming search requests, processes Battery ID search requests, matches the submitted Battery IDs with stored DBP records, and returns the corresponding DBP IDs	OAuth2.0 or eIDAS
Third party actor	Requesting actor	Initiates search requests. Submits Battery ID search requests and receives the matching DBP IDs	
Market surveillance authorities and other public bodies			

Ordinary Sequence

Actors authenticate and are authorized by the EO or by the MSP acting on behalf of the EO. Depending on the actor's role, authentication is performed using OAuth 2.0 for third party actors, or eIDAS for market surveillance authorities and other public bodies requiring a higher assurance level. After successful authentication and authorization, the actor receives an access token that encodes the role-based access rights required for the search request.

Using the access token, the actor sends a POST request to the ReadDPPIdsByProductIds endpoint. The request body contains the Product ID/ Battery ID parameter, which specifies one or more Battery IDs for which the matching DBP IDs are to be identified. Optionally, the actor may include a limit parameter to control the maximum number of results returned in the current response. If the actor continues a previously started paginated result listing, a cursor parameter may also be included. If used, the cursor must not be empty.

Upon receiving the request, the EO/MSP processes the submitted Battery IDs and returns a response containing the set of DBP IDs matching the submitted Battery IDs. This method returns only DBP IDs and not the passport content itself.

As an extension of this user story, the returned DBP IDs may be used in a subsequent controlled read process to retrieve the corresponding battery passport data, as described in User Story 6.2.2 "Reading Controlled Battery Passport Data".

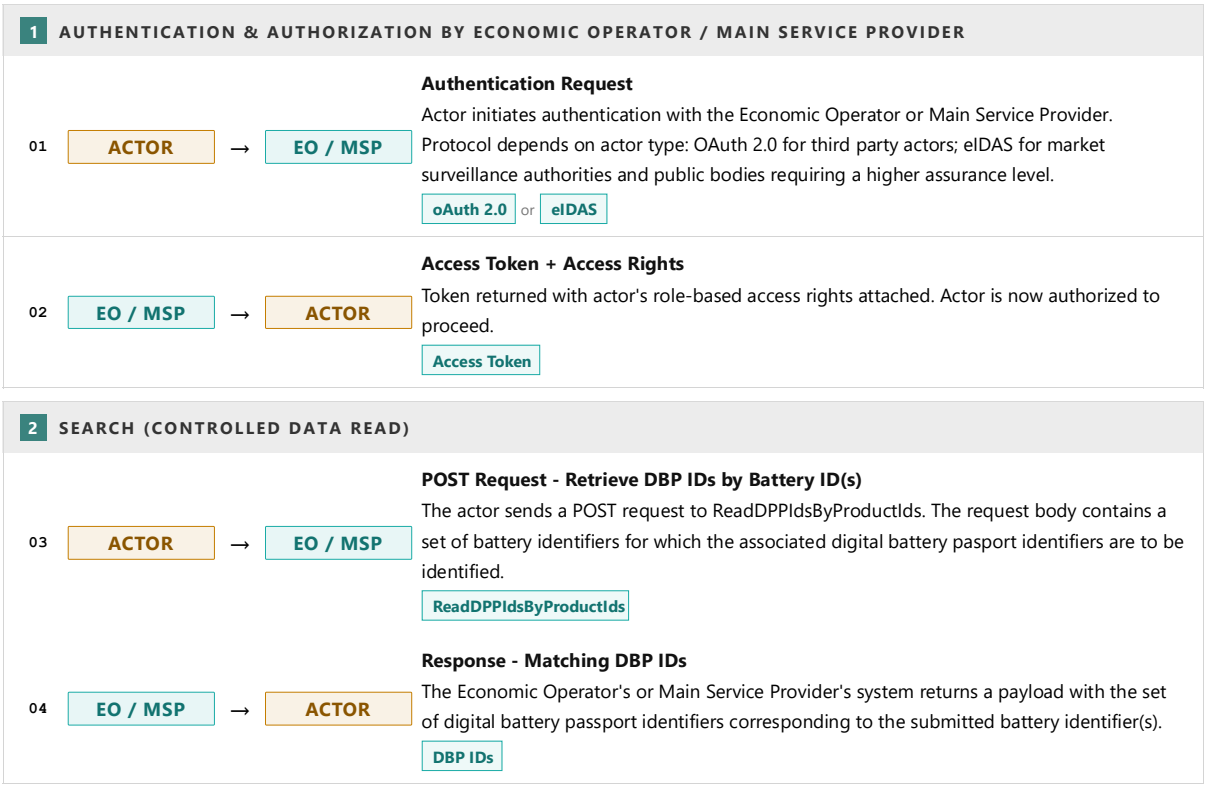


Figure 13: Ordinary Sequence "Search for Battery Passports"

Failure Scenarios

The following table summarizes the failure scenarios derived from the ordinary sequence of this user story. The ID column provides a unique reference for each failure scenario. The category column groups the scenarios according to the affected aspect of the process. The failure scenario column describes the deviation from the expected process flow that the DPP system must prevent or handle correctly.

ID	Category	Failure Scenario
FS-Input-001	Input Validation	Search request is processed although required input parameters are missing
FS-Input-002	Input Validation	Search request is processed although the Product ID (Battery ID) parameter is empty, malformed, or invalid
FS-Input-003	Input Validation	Search request is processed although the request body format is malformed or invalid
FS-Auth-001	Authentication	Search request is processed although the actor is not authenticated
FS-Auth-002	Authentication	EO/MSP accepts an invalid, expired, revoked, or malformed access token
FS-Auth-003	Authorization	EO/MSP grants an unauthorized actor access to the ReadDPPIdsByProductIds method
FS-Auth-004	Authorization / Access Control	Actor without search permission retrieves DBP IDs
FS-Acc-001	Access Control	Actor receives DPB IDs outside the actor's permitted discovery scope
FS-Acc-002	Access Control	Actor is denied DPB ID discovery although access rights are sufficient
FS-Search-001	Search	Valid search request returns no DPB IDs although matching DPBs exist
FS-Search-002	Search	Search request returns DPB IDs for the wrong Battery ID
FS-Search-003	Search	Search request returns an incomplete set of matching DPB IDs
FS-Search-004	Search	Search request returns duplicate DPB IDs in the response
FS-Search-005	Search	Search request returns DPB IDs that do not belong to any of the submitted Battery ID values
FS-Search-006	Search	Search request returns DPB content or additional restricted data instead of identifiers only
FS-Search-007	Search	Search request returns an error although no matching DPB IDs should result in an empty payload
FS-Pag-001	Pagination	Search request is processed although the cursor parameter is present but empty

ID	Category	Failure Scenario
FS-Pag-002	Pagination	Search request with a valid limit parameter is rejected without valid reason
FS-Pag-003	Pagination	Search request with a valid cursor parameter is rejected although pagination state exists
FS-Pag-004	Pagination	limit parameter is ignored and the response returns more results than requested
FS-Pag-005	Pagination	Pagination cursor resumes at the wrong position
FS-Pag-006	Pagination	Pagination returns overlapping results across pages
FS-Pag-007	Pagination	Pagination skips valid results between pages
FS-Pag-008	Pagination	Response omits a cursor although more results are available
FS-Pag-009	Pagination	Response returns a cursor although no more results are available
FS-Pag-010	Pagination	Search request is processed although the cursor is invalid, expired, or not linked to the actor's previous pagination state
FS-Scrape-001	Mass Retrieval Prevention	Search request is processed although the number of submitted Battery ID values exceeds the permitted request scope
FS-Scrape-002	Mass Retrieval Prevention	Actor is able to retrieve DPB IDs through automated or excessive repeated search requests without restriction
FS-Scrape-003	Mass Retrieval Prevention	Request frequency, result size, or pagination behavior exceeds the permitted search scope but is not restricted by the EO/MSP
FS-Scrape-004	Mass Retrieval Prevention	Search request is rejected although it remains within the permitted request size, result size, pagination, and frequency limits
FS-Resp-001	Response	Response does not follow the expected identifier list format
FS-Sys-001	System	Same search request produces inconsistent DPB ID results across repeated executions without data changes
FS-Sys-002	System / Audit Trail	Audit trail is missing or records the wrong actor, timestamp, request, submitted Battery ID, returned DPB IDs, or search outcome

Test Scenarios

The following table presents the test scenarios derived from the identified failure scenarios. The test scenario column names the case to be verified, while the objective column explains the purpose of the test. The preconditions column defines the conditions that must be fulfilled before the test is executed. The expected result column describes the system behavior required for a successful test outcome, and

the related failure scenario column links the test to the corresponding failure scenarios.

Test Scenario	Objective	Preconditions	Expected Result	Related Failure Scenario
Successful DBP ID discovery by Battery ID	Verify that a valid ReadDPPIdsBy-ProductIds request is processed successfully	Actor is authenticated and authorized; request body contains one or more valid Battery ID values; matching DBPs exist	Request succeeds; response contains the DBP IDs matching the submitted Battery ID values and follows the expected identifier list format	FS-Input-001, FS-Input-002, FS-Input-003, FS-Search-001, FS-Search-002, FS-Search-005, FS-Resp-001
Discovery result is complete, unique, and limited to identifiers only	Verify that the search function returns all matching DBP IDs exactly once and does not expose DBP content	Matching DBPs exist for the submitted Battery ID values; actor has sufficient discovery rights	Response contains the complete set of matching DBP IDs without duplicates; no DBP content or additional restricted data is returned	FS-Search-003, FS-Search-004, FS-Search-006
Actor receives only DBP IDs within the permitted discovery scope	Verify that discovery results are filtered according to the actor's access rights	Actor is authenticated and authorized with a defined discovery scope; matching DBPs exist inside and outside this scope	Response contains only DBP IDs within the actor's permitted discovery scope	FS-Auth-003, FS-Auth-004, FS-Acc-001, FS-Acc-002
Search with no matching DBP IDs returns an empty payload	Verify correct behavior when no DBP IDs match the submitted Battery ID values	Actor is authenticated and authorized; request contains valid Battery ID values, but no matching DBPs exist	Request succeeds; response contains an empty payload instead of an error	FS-Search-007
Pagination with valid limit and cursor works correctly	Verify that valid pagination parameters are accepted and applied correctly	Actor is authenticated and authorized; valid Battery ID values are submitted; result set spans multiple pages; valid limit and cursor are used	Response respects the limit; cursor resumes at the correct position; pages contain no overlapping or skipped results; cursor is returned only when further results are available	FS-Pag-002, FS-Pag-003, FS-Pag-004, FS-Pag-005, FS-Pag-006, FS-Pag-007, FS-Pag-008, FS-Pag-009
Search request with missing or	Verify that the search function validates	Actor is authenticated and authorized, but submits a	Request is rejected with a validation error;	FS-Input-001, FS-Input-002, FS-Input-003

Test Scenario	Objective	Preconditions	Expected Result	Related Failure Scenario
invalid input is rejected	mandatory input and request structure before processing	request with missing required parameters, empty or invalid Battery ID, or malformed request body	no search is performed and no DBP IDs are returned	
Unauthenticated or invalidly authenticated search request is rejected	Verify that the search function cannot be used without valid authentication	Actor sends a search request without authentication or with an invalid, expired, revoked, or malformed access token	Request is rejected; no DBP IDs are returned	FS-Auth-001, FS-Auth-002
Unauthorized actor or actor without search permission is rejected	Verify that only actors with the required authorization may use ReadDPPIdsBy-ProductIds	Actor is authenticated but not authorized to use the search method or does not have search permission	Request is rejected; no DBP IDs are returned	FS-Auth-003, FS-Auth-004
Search response contains identifiers only	Verify that the discovery method does not expose DBP content or additional restricted data	Actor is authenticated and authorized; valid search request is submitted; matching DBPs exist	Response contains only DBP IDs; no DBP content or additional restricted data is included	FS-Search-006
Request with empty, invalid, expired, or unrelated cursor is rejected	Verify that cursor validation is enforced for pagination	Actor is authenticated and authorized; request contains an empty cursor, invalid cursor, expired cursor, or cursor not linked to the actor's previous pagination state	Request is rejected with a validation error; pagination is not continued	FS-Pag-001, FS-Pag-010
Pagination returns non-overlapping and non-skipping results	Verify that pagination returns each matching DBP ID exactly once across pages	Search result spans multiple pages; actor uses valid limit and valid continuation cursors	Paginated responses contain no overlapping results and no valid results are skipped between pages	FS-Pag-006, FS-Pag-007
Search request exceeding permitted request scope is rejected or restricted	Verify that request size restrictions are enforced to prevent mass retrieval	Actor is authenticated and authorized, but submits more Battery ID values than permitted by the EO/MSP search policy	Request is rejected or restricted according to the permitted search scope	FS-Scrape-001

Test Scenario	Objective	Preconditions	Expected Result	Related Failure Scenario
Excessive repeated search requests are restricted	Verify that automated or excessive repeated search requests cannot be used for mass data retrieval	Actor repeatedly sends search requests at a frequency or volume exceeding the permitted search policy	EO/MSP restricts, throttles, or rejects excessive requests	FS-Scrape-002, FS-Scrape-003
Search request within permitted limits is not wrongly rejected	Verify that anti-scraping controls do not block valid requests	Actor is authenticated and authorized; request size, result size, pagination behavior, and request frequency remain within permitted limits	Request is accepted and processed normally	FS-Scrape-004
Response follows the expected identifier list format	Verify that the response payload follows the expected format for DBP ID discovery	Valid search request is processed successfully	Response contains the expected identifier list format and no malformed or unexpected payload structure	FS-Resp-001
Repeated identical search requests return consistent DBP ID results	Verify deterministic behavior when the same valid request is repeated without data changes	Actor is authenticated and authorized; matching DBP data remains unchanged; the same valid request is submitted multiple times	Each execution returns the same DBP ID result set	FS-Sys-001
Audit trail records the search request and outcome	Verify traceability of successful and rejected DBP ID discovery requests	Actor performs a successful or rejected ReadDPPIdsByProductIds request	Audit trail records the correct actor, timestamp, request, submitted Battery ID, returned DBP IDs if applicable, and search outcome	FS-Sys-002

Note: Some of the test scenarios described in this document may not be implemented within the scope of the test environment currently being developed. These scenarios are nevertheless included in this document to provide a complete overview of potential failure scenarios within the overall DPP system and to clarify their relevance for end-to-end system validation.

6.4 MAINTAIN BATTERY PASSPORT DATA

6.4.1 UPDATING OF BATTERY PASSPORT DATA

Overview

This user story describes how battery passport data is updated within the DBP system. It covers modifications to an existing battery passport, including updating one or multiple values, adding new data attributes, and deleting existing attributes. Updates may be triggered by lifecycle events, e.g., repair, refurbishment, ownership change or by data correction.

Assumptions

- The EO/MSP authenticates downstream actors working with the EO using OAuth 2.0.
- The BSP authenticates EO/MSP using OAuth 2.0.
- The EO/MSP authenticates regulated base roles using the eIDAS framework.
- The Create API between the EO/MSP and the BSP is used as a lifecycle API to create backup copies of the DBP after relevant updates.
- A backup copy of the existing DBP version has already been created by the BSP before an update is performed.
- Backup creation after an update is considered a technical measure to support data security, data restoration, and the availability of archived DBP versions.
- Backup creation after an update is included as a future-oriented approach in case the EO/MSP system becomes unavailable or the EO leaves the market.
- Only Recyclers and the EO may update the DBP status to Marked for Deletion.
- A DBP can only be updated if its current status is Active.
- If a DBP has the status Marked for Deletion or Archived, no further update functions may be executed and no new DBP ID may be created.
- The status Marked for Deletion is used to identify which DBPs may be deleted.
- The DBP ID is not a stable permanent identifier. It changes with each relevant update cycle, and each relevant update results in the creation of a new DBP ID.
- Frequently updated attributes do not require the creation of a new DBP version or a new DBP ID.

Actors and Roles

Actor	Role	Responsibilities	Authentication Protocol provided by
Economic Operator / Main Service Provider	DBP Data Provider	Hosts the DBP and manages access rights for all actors. Authenticates and authorizes incoming update requests.	OAuth 2.0 or eIDAS

Actor	Role	Responsibilities	Authentication Protocol provided by
Third Party Actor Regulated Base Role	Data Updater	Sends PATCH request with DBP ID and update payload. Must hold write rights for all requested attributes.	
Backup Service Provider	Storage Provider	Authenticates EO/MSP, creates new backup copy of updated DBP, archives previous backup, returns new DBP ID	OAuth 2.0

Ordinary Sequence

Actors authenticate and are authorized by the Economic Operator or by the Main Service Provider acting on behalf of the Economic Operator. Depending on the actor's role, authentication is performed using OAuth 2.0 for third party actors, or eIDAS for regulated base role with write permission. After successful authentication and authorization, the actor receives an access token that encodes the role-based access rights required for submitting update requests.

Using the access token, the actor submits a PATCH request to the EO/MSP, selecting the applicable update function and providing the required input parameters. The choice of function depends on the scope of the intended update. UpdateDPPById is used to update selected fields of a DBP identified by its DBP ID, whereas UpdateDataElement is used to update an individual data element identified by its path within the DBP. Upon receiving the request, the EO/MSP checks whether the actor has the necessary access rights for all data fields included in the update request. If access is granted, the update is applied by creating a new DBP instance with a new DBP ID. The new DBP becomes Active, while the previous DBP is archived. The system then returns the updated DBP content or the updated data element as confirmation that the update has been applied.

The update also triggers the creation of a new backup copy. For this purpose, the EO/MSP authenticates and is authorized by the Backup Service Provider using OAuth 2.0. After successful authentication and authorization, the EO/MSP sends the CreateDPP API call to the Backup Service Provider to create a backup copy of the new active DBP. The Backup Service Provider archives the previous backup copy, activates the new backup copy, and returns the DBP ID of the active version to the EO/MSP as confirmation. The receipt of this confirmation marks the successful completion of the DBP update process.

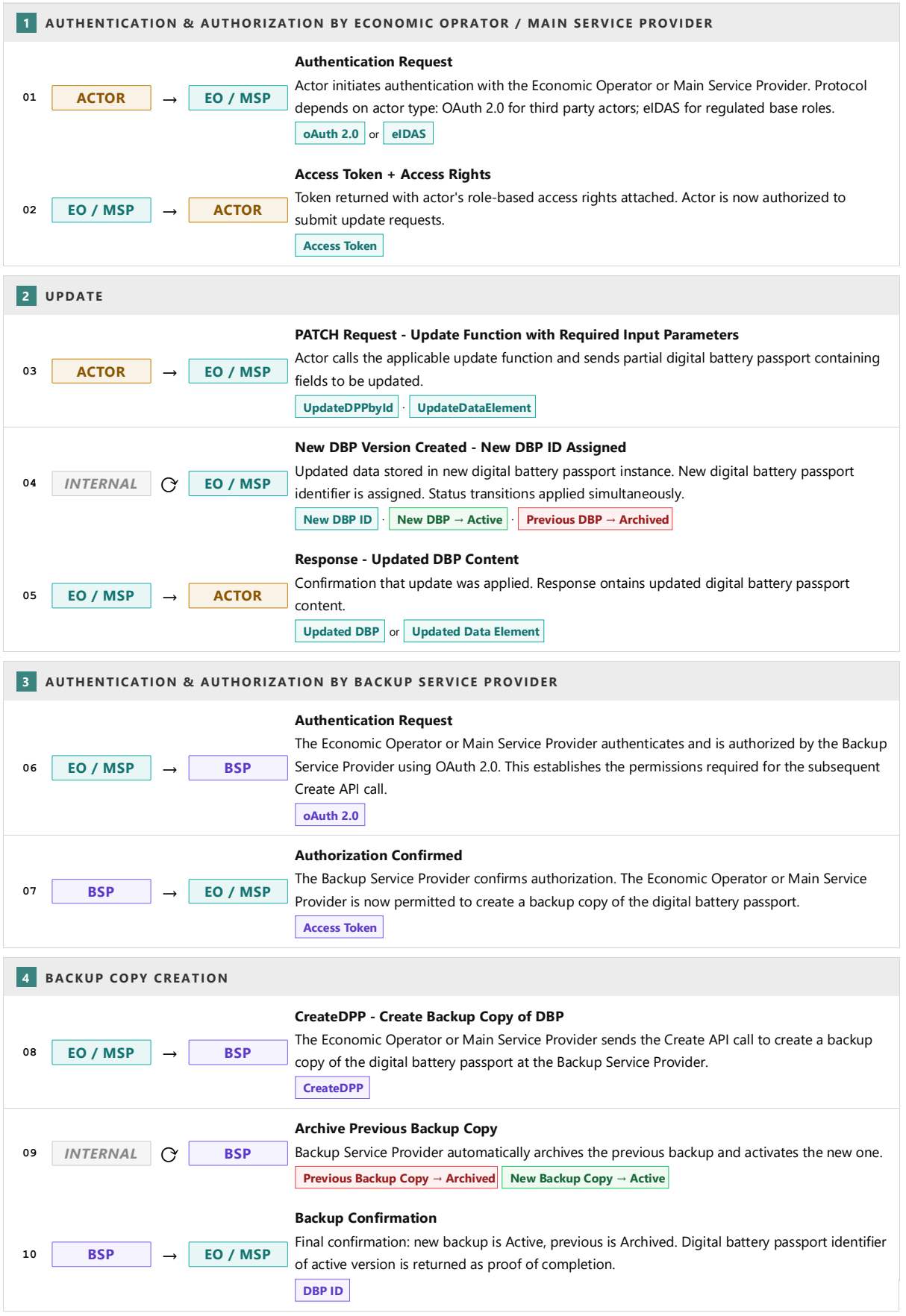


Figure 14: Ordinary Sequence "Updating of Battery Passport Data"

Failure Scenarios

The following table summarizes the failure scenarios derived from the ordinary sequence of this user story. The ID column provides a unique reference for each failure scenario. The category column groups the scenarios according to the affected aspect of the process. The failure scenario column describes the deviation from the expected process flow that the DPP system must prevent or handle correctly.

ID	Category	Failure Scenario
FS-Input-001	Input Validation	Update request is processed although required input parameters are missing
FS-Input-002	Input Validation	Update request is processed although the DBP ID, data element path, or update target is malformed or invalid
FS-Auth-001	Authentication	Update request is processed although the actor is not authenticated
FS-Auth-002	Authentication	DBP system accepts an invalid, expired, revoked, or malformed access token
FS-Auth-003	Authentication	DBP system grants an unauthorized actor write access to DBP data
FS-Auth-004	Authentication / Access Control	Actor without update rights updates a restricted data element
FS-Acc-001	Access Control	System updates data beyond the actor's access rights
FS-Acc-004	Access Control	updateDataElement updates a restricted element despite missing access rights
FS-Upd-001	Update	Valid update request is rejected despite sufficient permissions and valid input data
FS-Upd-002	Update	Invalid data format or data type is accepted
FS-Upd-003	Update / Status	DBP with status Archived or Marked for Deletion is updated although the update is not allowed
FS-Upd-004	Update	Same PATCH request is processed twice, creating duplicate update effects or duplicate DBP versions
FS-Upd-005	Update	API returns an error although the update was successfully applied
FS-Upd-006	Update	API returns success although the update was not applied
FS-Upd-007	Update	Update request modifies the wrong DBP or wrong data element
FS-Upd-008	Update	Valid addition of an allowed data element is rejected despite sufficient permissions
FS-Upd-009	Update	Valid deletion of an allowed data element is rejected despite sufficient permissions
FS-Upd-010	Update	Update request is processed although the referenced DBP does not exist

ID	Category	Failure Scenario
FS-Upd-011	Update	Update request is processed although the referenced data element does not exist
FS-Upd-012	Update / Schema Validation	Update request is processed although the payload contains fields outside the allowed DBP schema
FS-Upd-013	Update / Data Completeness	Update request removes mandatory DBP data although the resulting DBP becomes incomplete
FS-Upd-014	Update / Semantic Validation	Update request is accepted although updated values violate semantic rules, value ranges, or allowed units
FS-Upd-015	Update / Immutable Fields	Immutable identifiers or metadata, such as Product ID/ Battery ID, Economic Operator ID, or creation timestamp, are modified by an update request
FS-Atom-001	Update Atomicity	Partial changes are adopted in the DBP although one or more parts of the update request failed
FS-Atom-002	Update Atomicity	Update process does not fully fail although validation, authorization, or processing of at least one requested field fails
FS-Atom-003	Update Atomicity	New DBP version is created although the complete update request should have been rejected
FS-Atom-004	Update Atomicity	DBP is left in an inconsistent state with a mixture of old and updated values after a failed update request
FS-Atom-005	Update Atomicity / Backup	Backup copy is created for a partially failed update although no changes should have been adopted in the DBP
FS-Stat-001	Status	New DBP ID is created despite forbidden DBP status
FS-Stat-002	Status	Illegal status transition is performed
FS-UpdVer-001	Update Versioning	New DBP ID conflicts with an existing DBP ID
FS-UpdVer-002	Update Versioning	Two or more DBP versions remain active simultaneously after a successful update
FS-UpdVer-003	Update Versioning	No DBP version remains active after a successful update
FS-UpdVer-004	Update Versioning	Single update request produces multiple new DBP versions
FS-UpdVer-005	Update Versioning	Concurrent updates create versions in the wrong chronological order
FS-UpdVer-006	Update Versioning	Previous DBP version is not archived after the new DBP version becomes active
FS-UpdVer-007	Update Versioning	New DBP version is not correctly linked to the Battery ID
FS-UpdVer-008	Update Versioning	Current active DBP reference still points to the previous DBP version after a successful update
FS-UpdVer-009	Update Versioning	New DBP ID is not created although the update was successfully applied

ID	Category	Failure Scenario
FS-UpdVer-010	Update Versioning	New DBP version is created but assigned the wrong product ID
FS-UpdVer-011	Update Versioning	Previous and new DBP versions are both archived after a successful update
FS-UpdVer-012	Update Versioning	New DBP version is archived instead of active after a successful update
FS-BspAuth-001	BSP Authentication	BSP processes backup creation although the EO/MSP system is not authenticated via OAuth 2.0
FS-BspAuth-002	BSP Authentication	BSP accepts an invalid, expired, revoked, or malformed OAuth 2.0 token from the EO/MSP system
FS-BspAuth-003	BSP Authorization	Unauthorized EO/MSP system is able to create a backup copy
FS-Bck-001	Backup	Mandatory backup copy is not created after successful update
FS-Bck-002	Backup	BSP returns success although the backup copy was not actually created
FS-Bck-003	Backup	Wrong DBP dataset is stored as backup copy
FS-Bck-004	Backup	DBP ID is not returned after successful backup creation
FS-Bck-005	Backup	Wrong DBP ID is returned as backup confirmation
FS-Bck-006	Backup	Same backup creation request is processed twice, creating duplicate backup copies
FS-Bck-007	Backup Versioning	New backup copy is created, but previous backup copy is not archived
FS-Bck-008	Backup Versioning	Previous backup copy is archived, but new backup copy is not created
FS-Bck-009	Backup / Consistency	Backup failure leaves the DBP system and backup system in inconsistent states
FS-Bck-010	Backup	Backup system does not mark the new backup copy as active after successful update backup creation
FS-Bck-011	Backup Versioning	Two or more backup copies remain active simultaneously after successful update backup creation
FS-Bck-012	Backup Versioning	No backup copy remains active after successful update backup creation
FS-Bck-013	Backup Versioning	Previous and new backup copies are both archived after successful update backup creation
FS-Resp-001	Response	Updated DBP content is not returned although the update was successfully applied
FS-Resp-002	Response	Response contains outdated, incomplete, or wrong updated DBP content

ID	Category	Failure Scenario
FS-Resp-003	Response	New DBP ID is not returned although a new DBP version was created after a successful update
FS-Resp-004	Response	Response returns the wrong DBP ID or still returns the previous DBP ID after a successful update
FS-Sync-001	System / Synchronization	Synchronization lag causes different services to show different current DBP versions
FS-Sync-002	System / Synchronization	QR code scanning or read APIs still return the previous DBP version after a successful update
FS-Sys-001	System / Audit Trail	Audit trail is missing or records the wrong actor, timestamp, request, updated fields, previous DBP ID, new DBP ID, backup status, or update outcome
FS-Sys-002	System / Audit Trail	Rejected update request is not recorded in the audit trail although the rejection is security- or validation-relevant

Test Scenarios

The following table presents the test scenarios derived from the identified failure scenarios. The test scenario column names the case to be verified, while the objective column explains the purpose of the test. The preconditions column defines the conditions that must be fulfilled before the test is executed. The expected result column describes the system behavior required for a successful test outcome, and the related failure scenario column links the test to the corresponding failure scenarios.

Test Scenario	Objective	Preconditions	Expected Result	Related Failure Scenario
Successful single data element update	Verify that one valid data element can be updated successfully	DBP exists and is Active; actor is authenticated and authorized for the requested data element; update payload is valid	Update is accepted; exactly one new DBP version is created; new DBP ID is returned; previous DBP version is archived; new DBP version becomes active; updated DBP content is returned	FS-Upd-001, FS-UpdVer-001, FS-UpdVer-006, FS-UpdVer-007, FS-UpdVer-009, FS-Resp-001, FS-Resp-003
Successful multi-field update with full atomicity	Verify that multiple authorized fields can be updated in one all-or-nothing transaction	DBP is Active; actor is authenticated and authorized; actor has update rights for all requested fields; all fields are valid	All requested changes are applied in one transaction; exactly one new DBP version is created; no partial update occurs	FS-Atom-001, FS-Atom-002, FS-Atom-004
Successful addition of an allowed data element	Verify that an allowed new data element can be added	DBP is Active; actor is authenticated and authorized; actor has required rights;	New data element is stored in the new DBP version; previous DBP	FS-Upd-008

Test Scenario	Objective	Preconditions	Expected Result	Related Failure Scenario
		schema allows the new data element	version is archived; new DBP version is active	
Successful deletion of an allowed non-mandatory data element	Verify that an existing non-mandatory data element can be deleted	DBP is Active; data element exists; actor is authenticated and authorized; actor has delete/update rights; field is not mandatory	Data element is removed in the new DBP version; resulting DBP remains complete; previous DBP version is archived	FS-Upd-009, FS-Upd-013
Correct update response and DBP versioning after successful update	Verify that a successful update creates the correct new DBP version, returns the correct response, and maintains correct active/archive status	Valid update request is processed successfully	Exactly one new DBP version is created; new DBP ID is unique and returned; new version is active; previous version is archived; product reference points to the new active version; response contains complete updated DBP content and not outdated content	FS-UpdVer-001, FS-UpdVer-002, FS-UpdVer-003, FS-UpdVer-006, FS-UpdVer-007, FS-UpdVer-008, FS-UpdVer-009, FS-UpdVer-010, FS-UpdVer-011, FS-UpdVer-012, FS-Resp-001, FS-Resp-002, FS-Resp-003, FS-Resp-004
Concurrent updates are serialized correctly	Verify that concurrent update requests do not corrupt version history or create inconsistent active/archive states	Two authorized actors submit update requests for the same active DBP at nearly the same time	Updates are processed in a controlled chronological order; no version is silently overwritten; exactly one DBP version is active after each successful update	FS-UpdVer-004, FS-UpdVer-005
Successful backup creation, backup versioning, and backup idempotency after update	Verify that backup creation after update is complete, versioned correctly, and safe against duplicate requests	Update has been successfully applied; EO/MSP system is authenticated and authorized by BSP; backup creation request is submitted once and then repeated	Backup copy is created for the correct updated DBP dataset; BSP returns the correct DBP ID; previous backup copy is archived; new backup copy becomes active; repeated backup request does not create duplicate backup copies	FS-Bck-001, FS-Bck-002, FS-Bck-003, FS-Bck-004, FS-Bck-005, FS-Bck-006, FS-Bck-007, FS-Bck-008, FS-Bck-010, FS-Bck-011, FS-Bck-012, FS-Bck-013

Test Scenario	Objective	Preconditions	Expected Result	Related Failure Scenario
QR code scanning or read APIs return updated DBP version	Verify that QR code scanning or all read APIs expose the current DBP version after update	Update and backup have been completed successfully	QR code scanning or read APIs return the new active DBP version and not the previous version	FS-Sync-001, FS-Sync-002
Audit trail is created for successful and rejected update processes	Verify traceability of update processing and security-relevant rejections	Successful update is processed, or an invalid/security-relevant update request is rejected	Audit trail records correct actor, timestamp, request, updated fields, previous DBP ID, new DBP ID, backup status, rejection reason, and update outcome	FS-Sys-001, FS-Sys-002
Update request with missing or malformed input parameters is rejected	Verify that update APIs require valid input parameters	Actor submits update request with missing required input parameters, malformed DBP ID, invalid data element path, or invalid update target	Request is rejected; no update is applied; no new DBP version is created; no backup is created	FS-Input-001, FS-Input-002
Unauthenticated or invalidly authenticated update request is rejected	Verify that update requests require valid authentication	Actor submits update request without authentication or with invalid, expired, revoked, or malformed access token	Request is rejected; no update is processed; no new DBP version is created	FS-Auth-001, FS-Auth-002
Actor without sufficient update rights cannot update DBP data	Verify that write access is enforced for update operations	Actor is authenticated but has read-only rights or no update rights for the requested data element	Request is rejected; restricted data element is not updated; no partial update is applied	FS-Auth-003, FS-Auth-004, FS-Acc-001, FS-Acc-004
Update request with invalid data format or type is rejected	Verify validation of update payload values	DBP is Active; actor is authenticated and authorized; payload contains invalid data type or format	Request is rejected with validation error; no changes are adopted in the DBP	FS-Upd-002
Update request for wrong or non-existing target is rejected	Verify that the update target must exist and match the requested	Actor submits update request for a non-existing DBP, non-existing data element, wrong	Request is rejected; no DBP content is modified; no new DBP version is created	FS-Upd-007, FS-Upd-010, FS-Upd-011

Test Scenario	Objective	Preconditions	Expected Result	Related Failure Scenario
	DBP/data element	DBP, or wrong data element		
Update request with schema or semantic violations is rejected	Verify schema and semantic validation before update processing	Actor submits payload with fields outside the allowed DBP schema or values violating semantic rules, value ranges, or allowed units	Request is rejected; no changes are adopted in the DBP	FS-Upd-012, FS-Upd-014
Update request modifying immutable identifiers or metadata is rejected	Verify that immutable fields cannot be modified by update requests	Actor submits update request attempting to modify Battery ID, Economic Operator ID, creation timestamp, or other immutable metadata	Request is rejected; immutable data remains unchanged; no new invalid DBP version is created	FS-Upd-015
Update request removing mandatory DBP data is rejected	Verify that an update cannot make the DBP incomplete	Actor submits update request removing mandatory DBP data	Request is rejected; mandatory data remains unchanged; no incomplete DBP version is created	FS-Upd-013
Update on forbidden DBP status is rejected	Verify that DBPs with forbidden status cannot be updated	DBP status is Archived or Marked for Deletion; actor submits update request that is not allowed for this status	Request is rejected; no update is applied; no new DBP ID is created; no illegal status transition occurs	FS-Upd-003, FS-Stat-001, FS-Stat-002
Duplicate PATCH request does not create duplicate effects	Verify that repeated PATCH requests are rejected or handled idempotently	A valid PATCH request has already been processed; same request is submitted again	No duplicate update effect occurs; no duplicate DBP version is created; request is rejected or handled safely	FS-Upd-004, FS-UpdVer-004
Failed multi-field update does not adopt partial changes	Verify the all-or-nothing rule for update processing	Update request contains multiple fields; at least one field fails validation, authorization, or processing	Complete update fails; no field is changed; no partial DBP state is adopted	FS-Atom-001, FS-Atom-002, FS-Atom-004

Test Scenario	Objective	Preconditions	Expected Result	Related Failure Scenario
Failed update does not create new DBP version or backup copy	Verify that failed updates do not trigger versioning or backup creation	Update request fails because one or more parts of the request are invalid or unauthorized	No new DBP version is created; no backup copy is created; DBP remains unchanged	FS-Atom-003, FS-Atom-005
DBP version ID conflict is prevented	Verify uniqueness of the new DBP ID	Update request is successfully processed and system attempts to create new DBP ID	New DBP ID is unique and does not conflict with an existing DBP ID	FS-UpdVer-001
No missing or wrong new DBP ID after successful update	Verify that successful update always produces the correct new DBP ID	Update is successfully applied	New DBP ID is created, returned, and assigned to the correct updated DBP version and Battery ID	FS-UpdVer-009, FS-UpdVer-010, FS-Resp-003, FS-Resp-004
DBP active/archive status is consistent after successful update	Verify that update does not create invalid DBP version states	Valid update is successfully processed	Exactly one DBP version is active; previous and new versions are not both archived; new version is active, not archived	FS-UpdVer-002, FS-UpdVer-003, FS-UpdVer-011, FS-UpdVer-012
Previous DBP version is archived and active reference is updated	Verify correct transition from previous active version to new active version	Successful update has created a new DBP version	Previous DBP version is archived; current active reference points to the new version, not the previous version	FS-UpdVer-006, FS-UpdVer-008
Concurrent updates do not corrupt version order	Verify correct chronological ordering under parallel update requests	Two or more authorized actors update the same DBP concurrently	Updates are serialized or conflict-managed; chronological order is preserved; no version is silently overwritten	FS-UpdVer-005
Unauthenticated or invalidly authenticated BSP backup request is rejected	Verify that BSP backup creation requires valid OAuth 2.0 authentication	EO/MSP system calls BSP Create API without valid authentication or with invalid/expired/revoked/malformed OAuth 2.0 token	Request is rejected; no backup copy is created; no DBP ID is returned	FS-BspAuth-001, FS-BspAuth-002
Unauthorized EO/MSP system cannot create backup copy	Verify that only authorized EO/MSP systems can create backup copies	Actor is not authorized for backup creation but calls BSP Create API	Request is rejected; no backup copy is created	FS-BspAuth-003

Test Scenario	Objective	Preconditions	Expected Result	Related Failure Scenario
Backup creation failure is detected after successful update	Verify that missing or false backup creation is detected	DBP update succeeds; BSP does not create backup copy or returns success without actually creating it	System detects backup failure; inconsistency is reported; audit trail records the failure	FS-Bck-001, FS-Bck-002, FS-Bck-009
Backup copy must match the updated DBP dataset	Verify that backup copy contains the correct updated DBP content	Successful update is followed by backup creation	Backup copy corresponds to the correct updated DBP dataset and correct DBP ID	FS-Bck-003, FS-Bck-005
Backup confirmation must contain correct DBP ID	Verify correct backup confirmation handling	Backup copy is successfully created	BSP returns correct DBP ID as backup confirmation	FS-Bck-004, FS-Bck-005
Backup version status is consistent after successful update backup creation	Verify that backup active/archive logic is correct	Successful update backup creation is completed	Previous backup copy is archived; new backup copy is active; not both copies are archived; not more than one backup copy is active; at least one backup copy remains active	FS-Bck-007, FS-Bck-008, FS-Bck-010, FS-Bck-011, FS-Bck-012, FS-Bck-013
Duplicate backup creation request does not create duplicate backups	Verify idempotency of backup creation	Same backup creation request is submitted more than once	Only one backup copy is created, or repeated request is handled safely according to idempotency rules	FS-Bck-006
System synchronization returns the correct current version	Verify consistency across services after update	Successful update and backup creation have been completed	All relevant services and read APIs show the same current DBP version	FS-Sync-001, FS-Sync-002
Audit trail records rejected update requests	Verify traceability of security- or validation-relevant rejected requests	Update request is rejected due to authentication, authorization, validation, status, or atomicity failure	Rejected request is recorded with actor, timestamp, request, rejection reason, and outcome	FS-Sys-002

Note: Some of the test scenarios described in this document may not be implemented within the scope of the test environment currently being developed. These scenarios are nevertheless included in this document to provide a complete overview of potential failure scenarios within the overall DPP system and to clarify their relevance for end-to-end system validation.

6.4.2 DELETION OF BATTERY PASSPORT

Overview

This user story describes how a Digital Product Passport is removed from public access and active hosting within the DPP system. Deletion is a terminal operation – it removes the active battery passport from the Economic Operator's server and the Backup Service Provider's system.

Assumptions

- The BSP authenticates the EO and the MSP using OAuth 2.0.
- EO/MSP authentication and authorization are performed by the EC Registry using the eIDAS framework.
- The status Marked for Deletion is used to identify which DBPs may be deleted.
- A DBP may be deleted only if its current version has the status Marked for Deletion.
- Once the DBP status has been set to Marked for Deletion, no further update functions may be executed and no new DBP ID may be created.
- Only the EO/MSP may invoke the DELETE API.
- Deletion applies to all DBP versions linked to the Battery ID and to the corresponding backup copies stored by the Backup Service Provider.
- After successful deletion, the DBP is no longer accessible via QR code and Read APIs.

Actors and Roles

Actor	Role	Responsibilities	Authentication Protocol provided by
Economic Operator / Main Service Provider	DPP Owner / DBP Data Provider	Holds the authority to delete the DBP, checks the deletion preconditions, requests deletion from the BSP, stops public hosting, notifies the EC Registry	
Backup Service Provider	Storage Provider	Authenticates and authorizes the EO, deletes the marked-for deletion backup instance	OAuth 2.0
EC Registry	Registration Authority	Receives the deletion notification from the EO and retains the necessary registration information.	eIDAS

Ordinary Sequence

Before deletion can be initiated, the DBP must already have been set to the status Marked for Deletion through an update process (see Chapter 6.3.1). Once this status has been assigned, the EO/MSP acting on its behalf treats the passport as frozen. No

further update functions may be executed and no new DBP ID may be created for this passport.

The EO/MSP first authenticates and is authorized by the BSP using OAuth 2.0. After successful authentication and authorization, the EO/MSP sends a DeleteDPPbyId request to the BSP using the DBP ID as the input parameter. The BSP verifies the authorization and deletes the marked-for-deletion backup instance according to the defined deletion and retention rules. After successful deletion, the BSP returns an HTTP 200 OK confirmation to the EO/MSP.

After receiving the deletion confirmation, the EO/MSP removes the corresponding Marked for Deletion version from its own server. As a result, no active DBP version remains in the operational DBP system. Public hosting of the DBP ends, QR-code access is terminated, and regular read functions can no longer retrieve the deleted passport.

Finally, the EO/MSP authenticates and is authorized by the EC Registry using eIDAS. After authorization has been confirmed, the EO/MSP notifies the EC Registry that the DBP has been deleted. This deletion notification allows the registry to maintain the necessary registration information for traceability while reflecting that the active battery passport is no longer available.

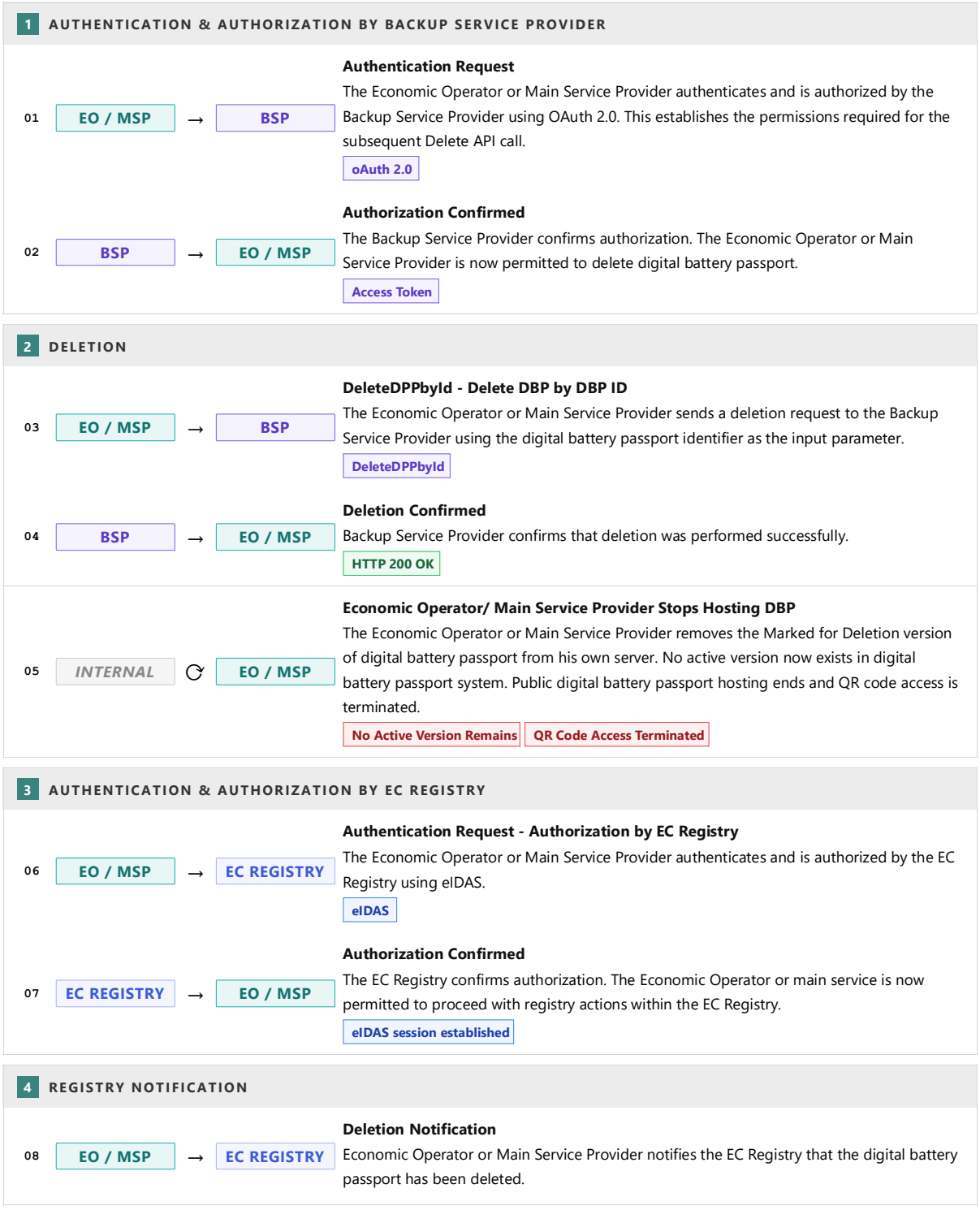


Figure 15: Ordinary Sequence "Deletion of Battery Passport"

Failure Scenarios

The following table summarizes the failure scenarios derived from the ordinary sequence of this user story. The ID column provides a unique reference for each failure scenario. The category column groups the scenarios according to the affected aspect of the process. The failure scenario column describes the deviation from the expected process flow that the DPP system must prevent or handle correctly.

ID	Category	Failure Scenario
FS-Input-001	Input Validation	Delete request is processed although required input parameters are missing
FS-Input-002	Input Validation	Delete request is processed although the DBP ID is malformed or invalid
FS-Auth-001	Authentication	Delete request is processed although the EO/MSP is not authenticated
FS-Auth-002	Authentication	BSP accepts an invalid, expired, revoked, or malformed OAuth 2.0 token
FS-Auth-003	Authentication	BSP grants an unauthorized EO/MSP delete access to DBP data
FS-Auth-004	Authentication / Access Control	Actor without delete rights deletes a DBP or backup copy
FS-Auth-005	Authorization / Tenant Boundary	Actor deletes DBP or backup copies belonging to another EO/MSP
FS-Pre-001	Preconditions	Delete request is processed although the DBP status is not Marked for Deletion
FS-Del-001	Deletion	Valid delete request is rejected although all preconditions are fulfilled
FS-Del-002	Deletion	Wrong DBP instance or wrong backup instance is deleted due to incorrect DBP ID resolution
FS-Del-003	Deletion	BSP returns success although deletion was not actually performed
FS-Del-004	Deletion	Same delete request is processed twice, causing duplicate side effects or inconsistent downstream behavior
FS-Del-005	Deletion	Delete request is completed although not all DBP versions linked to the Battery ID and their corresponding backup copies have been deleted
FS-Del-006	Deletion	Delete request is processed although the referenced DBP or backup copy does not exist
FS-Del-007	Deletion	Delete request deletes the wrong DBP
FS-Host-002	Hosting / Access	BSP confirms deletion, but EO/MSP fails to remove the local Marked for Deletion version and stop hosting
FS-Host-003	Hosting / Access	Public hosting of the DBP remains available after deletion has been completed
FS-Host-004	Hosting / Access	QR code access or read APIs still return DBP content linked to the Battery ID after deletion
FS-RegAuth-001	Registry Authentication	EC Registry processes a Registry action although the EO/MSP is not authenticated via eIDAS
FS-RegAuth-002	Registry Authentication	EC Registry accepts invalid, expired, revoked, or malformed eIDAS authentication for the EO/MSP

ID	Category	Failure Scenario
FS-RegAuth-003	Registry Authorization	EC Registry grants Registry access to an actor who is not authorized to act as the EO
FS-RegVer-001	Registry Verification	EC Registry allows an authenticated but unverified EO/MSP to perform Registry actions
FS-RegVer-002	Registry Verification	EC Registry accepts EO/MSP identity verification although the provided qualified e-signature, e-seal, eID, or attestation is missing or invalid
FS-RegVer-003	Registry Verification	EC Registry rejects EO/MSP identity verification although valid verification evidence is provided
FS-RegVer-004	Registry Verification	EC Registry does not recognize the EO/MSP as verified although the verified status is still within its valid three-year period
FS-RegVer-005	Registry Verification	EC Registry still treats the EO/MSP as verified although the verified status has expired
FS-RegProf-001	Registry Profile	EO/MSP profile data is not correctly stored, updated, or associated with the verified EO/MSP
FS-RegNot-001	Registry Notification	Deletion is completed, but the EC Registry is not notified
FS-RegNot-002	Registry Notification	Registry notification contains a wrong DBP ID or Battery ID
FS-RegNot-003	Registry Notification	EC Registry is notified before deletion has actually been completed
FS-RegNot-004	Registry Notification	EC Registry accepts a deletion notification although the referenced DBP has not been deleted
FS-Stat-001	Status	After deletion, an active or publicly accessible DBP version linked to the Battery ID still remains on the EO/MSP or BSP side
FS-Resp-001	Response	Deletion confirmation is not returned although deletion was successfully performed
FS-Sync-001	System / Synchronization	Synchronization lag causes EO/MSP, BSP, or EC Registry to show different deletion states
FS-Sys-001	System / Consistency	Partial failure leaves EO/MSP, BSP, and EC Registry in inconsistent states
FS-Sys-002	System / Audit Trail	Audit trail is missing or records the wrong actor, timestamp, request, DBP ID, Battery ID, deletion step, system outcome, or registry notification status

Test Scenarios

The following table presents the test scenarios derived from the identified failure scenarios. The test scenario column names the case to be verified, while the objective column explains the purpose of the test. The preconditions column defines the conditions that must be fulfilled before the test is executed. The expected result column describes the system behavior required for a successful test outcome, and

the related failure scenario column links the test to the corresponding failure scenarios.

Test Scenario	Objective	Preconditions	Expected Result	Related Failure Scenario
Successful end-to-end deletion of all DPB versions linked to the Battery ID	Verify that the complete delete workflow is executed correctly when all preconditions are fulfilled	DPB status is Marked for Deletion; EO/MSP is authenticated and authorized; all DPB versions linked to the Battery ID and corresponding backup copies exist; EO/MSP is verified for Registry actions	BSP deletes all relevant backup copies linked to the Battery ID; EO/MSP removes local DPB versions and stops hosting; QR code access and read APIs no longer return DPB content; EC Registry is notified; audit trail is created	FS-Del-001, FS-Del-002, FS-Del-005, FS-Host-002, FS-Host-003, FS-Host-004, FS-RegNot-001, FS-Stat-001, FS-Resp-001, FS-Sync-001, FS-Sys-001, FS-Sys-002
Delete request with missing or malformed input is rejected	Verify that deletion cannot be processed without valid required input parameters	Delete request is submitted with missing required parameters, malformed DPB ID	Delete request is rejected; no DPB version or backup copy is deleted; no Registry notification is sent	FS-Input-001, FS-Input-002
Unauthenticated or invalidly authenticated delete request is rejected by BSP	Verify that BSP does not process delete requests without valid OAuth 2.0 authentication	EO/MSP sends delete request without authentication or with invalid, expired, revoked, or malformed OAuth 2.0 token	Request is rejected; deletion is not performed; no success response is returned	FS-Auth-001, FS-Auth-002
Unauthorized actor cannot delete DPB data or backup copies	Verify that only an authorized EO/MSP with delete rights can initiate deletion	Actor is authenticated but does not have delete rights, or actor is not authorized for the specific DPB or backup copy	Delete request is rejected; no DPB data or backup copy is deleted	FS-Auth-003, FS-Auth-004
Cross-tenant deletion is prevented	Verify that an EO/MSP cannot delete DPB data or backup copies belonging to another EO/MSP	Actor submits delete request for a DPB or backup copy belonging to another EO/MSP	Delete request is rejected; no cross-tenant data is deleted	FS-Auth-005
Deletion is rejected if DPB is not marked for deletion	Verify that deletion can only start when the DPB has the required terminal status	DPB exists, but its current status is not Marked for Deletion	Delete request is rejected; no backup deletion, no local deletion, and no Registry notification are performed	FS-Pre-001
Valid delete request is not rejected when all	Verify that a valid delete request	DPB status is Marked for Deletion; EO/MSP is	Delete request is accepted and the	FS-Del-001

Test Scenario	Objective	Preconditions	Expected Result	Related Failure Scenario
preconditions are fulfilled	can be accepted and processed	authenticated and authorized; DPB ID is valid; related backup copies exist	deletion workflow starts	
Correct DPB and backup instances are selected for deletion	Verify that DPB ID resolution identifies the correct DPB and corresponding backup copies	Valid delete request is submitted with a DPB ID linked to a specific Battery ID	Only the intended DPB versions and corresponding backup copies are selected for deletion	FS-Del-002, FS-Del-007
Delete request for non-existing DPB or backup copy is rejected	Verify that deletion is not processed when the referenced object does not exist	Delete request references a DPB or backup copy that does not exist	Request is rejected; no deletion is performed; error response is returned	FS-Del-006
All DPB versions linked to the Battery ID are deleted	Verify that deletion is not limited to one single DPB version when multiple versions are linked to the same Battery ID	Battery ID is linked to several DPB versions and corresponding backup copies	All DPB versions linked to the Battery ID and all corresponding backup copies are deleted or made unavailable according to the deletion logic	FS-Del-005, FS-Stat-001
BSP returns success only after deletion was actually performed	Verify that a success response is only returned after actual deletion	Valid delete request is submitted and BSP deletion processing is monitored	BSP returns success only if deletion was actually performed; no false positive success response is returned	FS-Del-003, FS-Resp-001
Duplicate delete request is rejected or handled idempotently	Verify that repeated delete requests do not create duplicate side effects or inconsistent system states	A valid delete request has already been processed; the same request is submitted again	Duplicate request is rejected or handled idempotently; no duplicate side effects occur	FS-Del-004
EO/MSP removes local DPB versions and stops hosting after BSP confirmation	Verify correct follow-up processing on the EO/MSP side after successful BSP deletion	BSP has successfully completed deletion and returned confirmation	EO/MSP removes the local Marked for Deletion version, stops public hosting, and no publicly accessible DPB version remains	FS-Host-002, FS-Host-003, FS-Stat-001
QR code access and read APIs no longer return	Verify that deleted DPB content is no longer publicly accessible	Deletion workflow has been completed successfully	QR code access and read APIs no longer return DPB	FS-Host-004, FS-Stat-001

Test Scenario	Objective	Preconditions	Expected Result	Related Failure Scenario
DPB content after deletion			content linked to the Battery ID	
Registry action is rejected without valid eIDAS authentication	Verify that EC Registry actions require valid eIDAS authentication	EO/MSP submits Registry action without eIDAS authentication or with invalid, expired, revoked, or malformed eIDAS authentication	EC Registry rejects the action; no Registry deletion notification is accepted	FS-RegAuth-001, FS-RegAuth-002
Unauthorized actor cannot perform Registry action on behalf of the EO	Verify that only an actor authorized to act as the EO can perform Registry actions	Actor is authenticated but not authorized to act as the EO	EC Registry rejects the Registry action	FS-RegAuth-003
Registry action is rejected if EO/MSP is not verified	Verify that an authenticated but unverified EO/MSP cannot perform Registry actions	EO/MSP is authenticated via eIDAS but has no valid Registry verification status	EC Registry rejects the Registry action	FS-RegVer-001
Registry verification evidence is validated correctly	Verify that EC Registry accepts only valid verification evidence	EO/MSP submits qualified e-signature, e-seal, eID, or attestation for verification	Invalid or missing evidence is rejected; valid evidence is accepted	FS-RegVer-002, FS-RegVer-003
Three-year Registry verification validity period is enforced	Verify that the EC Registry correctly handles the validity period of EO/MSP verification	EO/MSP verification status exists and is either still within or already outside the valid three-year period	EO/MSP is recognized as verified within the valid three-year period; expired verification is no longer accepted	FS-RegVer-004, FS-RegVer-005
EO/MSP Registry profile is correctly linked to the verified actor	Verify that Registry profile data is correctly stored and associated with the verified EO/MSP	EO/MSP is authenticated and verified; Registry profile data exists or is updated	Profile data is correctly stored, updated, and associated with the verified EO/MSP	FS-RegProf-001
EC Registry is notified only after deletion is completed	Verify that Registry notification is sent only after the deletion workflow has actually been completed	BSP deletion and EO/MSP local deletion are executed; workflow completion status is known	EC Registry is notified after deletion completion only; no premature notification is sent	FS-RegNot-001, FS-RegNot-003
Registry notification contains correct DPB ID and Battery ID	Verify correctness of the deletion notification sent to the EC Registry	Deletion has been completed and Registry notification is triggered	Registry notification contains the correct DPB ID and Battery ID	FS-RegNot-002

Test Scenario	Objective	Preconditions	Expected Result	Related Failure Scenario
EC Registry rejects notification for a DPB that has not been deleted	Verify that EC Registry does not accept deletion notification for a non-deleted DPB	Registry notification references a DPB for which deletion has not actually been completed	EC Registry rejects or does not finalize the deletion notification	FS-RegNot-004
Cross-system deletion state is synchronized correctly	Verify that EO/MSP, BSP, and EC Registry show the same deletion state	Deletion workflow is completed across EO/MSP, BSP, and EC Registry	All systems show a consistent deletion state; no synchronization lag leads to contradictory states	FS-Sync-001, FS-Sys-001
Partial deletion failure does not leave inconsistent system states	Verify that partial failure is detected and handled consistently	Failure is injected during BSP deletion, EO/MSP local deletion, or Registry notification	System detects the partial failure and prevents or flags inconsistent final states across EO/MSP, BSP, and EC Registry	FS-Sys-001
Audit trail records the complete deletion workflow	Verify traceability of successful and failed deletion workflows	Delete workflow is executed successfully or with an injected failure	Audit trail records correct actor, timestamp, request, DPB ID, Battery ID, deletion step, system outcome, and Registry notification status	FS-Sys-002

Note: Some of the test scenarios described in this document may not be implemented within the scope of the test environment currently being developed. These scenarios are nevertheless included in this document to provide a complete overview of potential failure scenarios within the overall DPP system and to clarify their relevance for end-to-end system validation.

6.5 MANAGE END-OF-LIFE AND CIRCULARITY EVENTS REPAIR AND RECYCLING ACTIVITIES

Overview

This user story illustrates how existing DBP interaction modules can be combined in a circularity-related lifecycle scenario. Using the example of a Repairer or Recycler, it shows how public access, controlled data access, and update functions may support repair or recycling activities within the DBP ecosystem.

The purpose of this user story is not to introduce a separate technical function, but to demonstrate how previously described modules can be reused to document lifecycle-relevant changes after a physical repair or recycling activity. The physical activity itself and internal follow-up processes of the EO or MSP, such as backup creation, are outside the scope of this scenario.

Actors and Roles

Actor	Role	Responsibilities	Authentication Protocol provided by
Economic Operator / Main Service Provider	DPP Owner / DBP Data Provider	Hosts the DBP system, controls access to DBP data, validates submitted updates, and returns the updated content or confirmation response.	OAuth 2.0 or eIDAS
Repairer / Recycler	Circularity Actor	Performs repair or recycling activities on batteries, accesses the required DBP data within the permitted scope, and submits updates to document the new battery state.	

Ordinary Sequence

The process may begin with an optional QR code read. If needed, the Repairer or Recycler scans the QR code attached to the battery using a mobile device. The device extracts the URI encoded in the QR code and sends a secure internet request to the EO or MSP server. Since this step concerns publicly accessible DBP data, no authentication is required.

The EO or MSP returns the public battery passport data in a structured, human-readable HTML format. This allows the actor to review public battery information before requesting access to controlled data.

If additional controlled information is required for the intended repair or recycling activity, the actor authenticates with the EO or MSP. Depending on the actor type and the applicable identity management framework, authentication may be performed using OAuth 2.0 or eIDAS. After successful authentication and authorization, the actor receives an access token with the permitted read and write rights.

Using the access token, the Repairer or Recycler submits a GET request to retrieve the controlled DBP data required for the intended activity. Depending on the used

API function and input parameters, the request may refer to a DBP ID, Battery ID, or a specific data element path.

The EO or MSP checks the actor's access rights and returns only the data that the actor is entitled to access.

Based on the retrieved information, the Repairer or Recycler carries out the physical repair or recycling activity. This operational activity itself is outside the technical scope of the DBP system. From the system perspective, the relevant next step is the documentation of the resulting lifecycle change.

After completing the activity, the actor submits an update request to the EO or MSP. The request contains the DBP data elements to be updated and must remain within the actor's permitted write scope. Depending on the applicable update function, the actor may update selected fields of a DBP by its DBP ID or update a specific data element by its path within the DBP.

The EO or MSP validates the request and verifies whether the actor is permitted to update the addressed data elements. If the update is accepted, the system processes the submitted changes and returns the updated DBP content or the updated data element as confirmation.

As an optional final step, the Repairer or Recycler may submit a second controlled read request to verify that the update has been correctly stored and is reflected in the DBP. The EO or MSP returns the current DBP data within the actor's access scope, including the updated values from the previous update step.

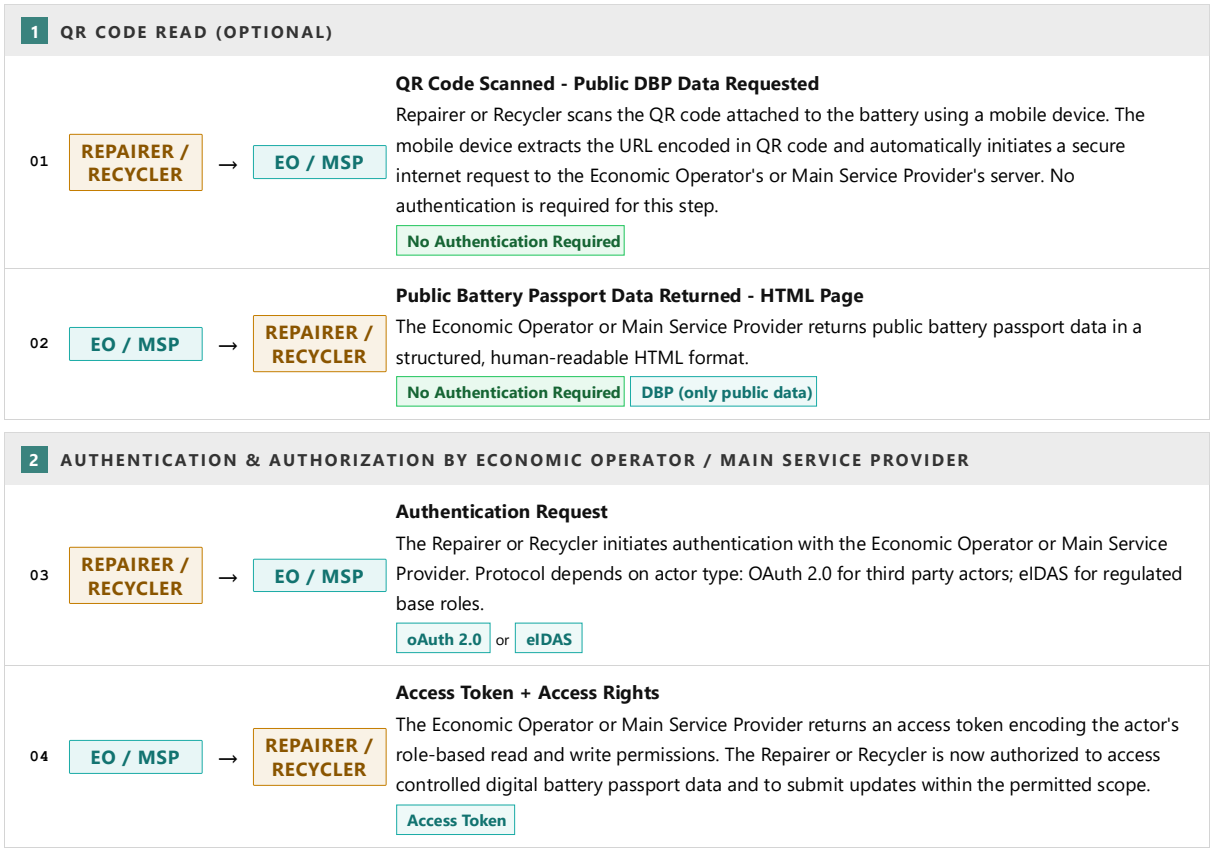


Figure 16: Ordinary Sequence "End-of-Life and Circularity Events. Part 1"

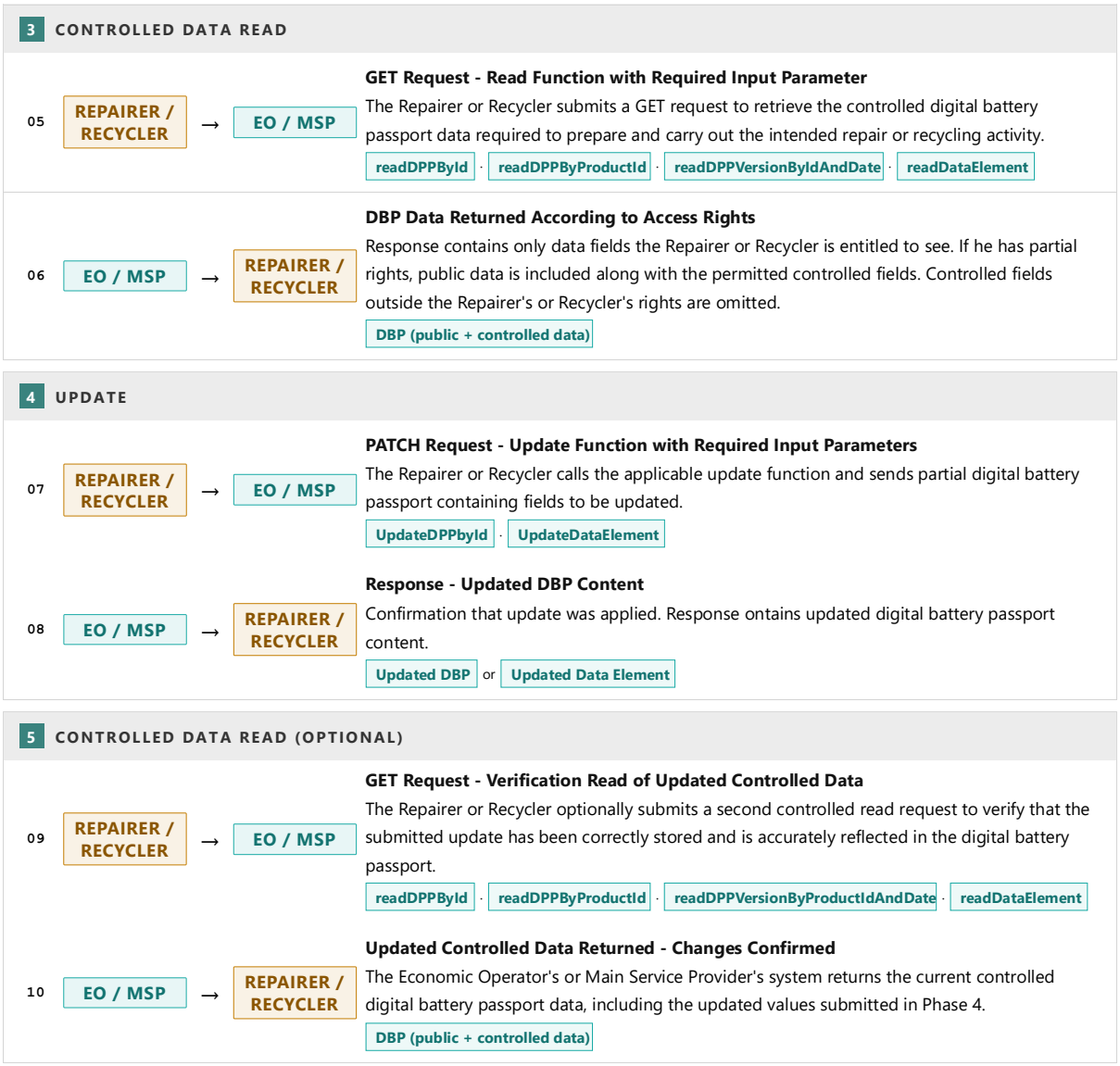


Figure 17: Ordinary Sequence "End-of-Life and Circularity Events. Part 2"

Scope Delimitation

This user story is included to demonstrate how previously defined modular user stories can be combined into a broader lifecycle scenario. It reuses interaction patterns for public data access, controlled data access, authentication and authorization, and data update.

For this reason, repair and recycling activities are not treated as a separate test focus in the initial test environment. No dedicated assumptions, failure scenarios, or test cases are defined specifically for this combined user story. Instead, the assumptions made for the corresponding previous user stories also apply here.

Failures may still occur within each reused module, for example during QR code access, authentication, controlled data retrieval, or data update. These failures are therefore module-specific rather than specific to the repair or recycling scenario and are covered by the respective previous user stories.

7 OUTLOOK

This document, *Navigating the Digital Battery Passport: User Stories from a Technical Perspective*, presents the first version of a user-story-based technical perspective on the battery passport ecosystem developed by the BatteryPass-Ready consortium. It was prepared while the underlying regulatory and standardization framework was still evolving.

The release of the second version is linked to the availability of the secondary legal acts and standards that will further specify processes that are not yet fully defined. These include, in particular, the Registry Act, with Commission adoption planned for Q3 2026; the Access Rights Act, which will address access rights to specific parts of the battery passport; and the Service Provider Act, which will define rules for service providers, with Commission adoption planned for Q4 2026. In addition, the CEN/CENELEC JTC 24 standards EN 18239, covering access rights management, information system security, and business confidentiality, and EN 18246, covering data authentication, reliability, and integrity, are currently still in draft form and are expected to be published at the end of Q3 2026.

Once these outstanding acts and standards are in place, the second version of this document will extend the scope and refine the existing user stories to reflect the consolidated requirements.

The identifiers addressed throughout the user stories are based on the BatteryPass-Ready Data Attribute Longlist v1.3. The next revision of the Attribute Longlist is planned for July or August 2026, following the adoption of the Registry Act. This revision will incorporate the resulting clarifications and ensure that the set of attributes, their classifications, and their access conditions remain aligned with the current regulatory requirements.

The test environment is based on failure- and test scenarios defined throughout this document and will be maintained in continuous alignment with them. As the user stories are extended and refined in line with the developments described above, the test environment will be updated accordingly so that it continues to reflect a subset of the scenarios described.

The BatteryPass-Ready consortium welcomes feedback on this document and the user stories presented herein. Comments, questions, and suggestions may be directed to the consortium or directly to Elena Andrushchenko and Dogan Efe, either by email or through exchange sessions.

8 REFERENCES

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